wellcome connecting science

Clinical validation/calibration of MAVEs

Miranda Durkie, Charlie Rowlands, Sophie Allen

November 2025

Session Agenda

1. Intro to ACMG v3/v4 and ACGS guidelines (Miranda Durkie)
2. Weighting functional evidence: Past and Present
    - Historical application of functional evidence in variant classification (Sophie Allen)
    - Bayesian conversion using LLR/Exponents (Sophie Allen)
    - Introduction to variant ‘truthsets’ (Charlie Rowlands)

-- Practice session 1: Defining variant truthsets –

- The Brnich et al Methodology (Charlie Rowlands)

-- Practice session 2: Assessing evidence weighting --

3. Weighting functional evidence: The future
    - Variant-level evidence weighting (Sophie Allen)
    - Combining MAVEs for clinical classification (Charlie Rowlands)

Session Objectives

- Objective 1: Understand the **ACMG/ACGS guidelines** currently in use to classify variants
- Objective 2: Understand and utilise the **points-based evidence system (SVC Tavtigian)** to weight functional assay data
- Objective 3: Apply the **ACMG v3.0 SVC Brnich methodology** to assign weighting to a functional assay
- Objective 4: Learn about new and upcoming methodologies used for **variant-level weighting**
- Objective 5: Learn how to **combine data from multiple assays**

Objective 1: Understand the **ACMG/ACGS guidelines** currently in use to classify variants

Historical application of functional evidence in variant classification

Sophie Allen

November 2025

Historic application of functional evidence

Inconsistent weighting as evidence

Rarely calibrated against robust clinical classifications

Limited guidance on number/performance of controls

Number of controls? **Type of variant?**

Current practice

General guidelines produced to assist clinical scientists/variant curators in using the data

Functional evidence weighted using the points-based evidence system (**SVC Tavtigian**)

Curation of **variant truthsets**

Clearer methodology (**SVC Brnich**) has been defined for clinical teams and expert groups to assess if a functional assay can be used in classification

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© American College of Medical Genetics and Genomics ACMG STANDARDS AND GUIDELINES | Genetics inMedicine

Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology

Sue Richards, PhD[1], Nazneen Aziz, PhD[2,16], Sherri Bale, PhD[3], David Bick, MD[4], Soma Das, PhD[5], Julie Gastier-Foster, PhD[6,7,8], Wayne W. Grody, MD, PhD[9,10,11], Madhuri Hegde, PhD[12], Elaine Lyon, PhD[13], Elaine Spector, PhD[14], Karl Voelkerding, MD[13] and Heidi L. Rehm, PhD[15]; on behalf of the ACMG Laboratory Quality Assurance Committee

Key principles:

- Functional data = ‘strong’ evidence
- Not all functional studies are created equally
- Need to consider how closely the assay reflects biological environment
- Consider: reproducibility, robustness, and validation data
- Assays looking at mRNA can be highly informative for splice variant effect

2015
ACMG v3.0

Richards et al., 2015, Genetics in Medicine

General guidelines produced to assist clinical scientists/variant curators in using the data

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- Consider: reproducibility, robustness, and validation data
- Assays looking at mRNA can be highly informative for splice variant effect

New general framework!

2015
ACMG v3.0

2026
ACMG v4.0

Current practice

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Objective 2: Understand and utilise the **points-based evidence system (SVC Tavtigian)** to weight functional assay data

Bayesian conversion using LLR/Exponents

Sophie Allen

November 2025

ACMG v3 Weightings

	Evidence Strength
Evidence of pathogenicity	P: Very Strong
	P: Strong
	P: Moderate
	P: Supporting
Evidence of benignity	B: Supporting
	B: Strong

Very confident evidence = pathogenicity (ie variant which deletes several exons in a region we know is essential to protein function)

Very confident evidence = benignity (ie variant which is very common in the general population)

ACMG v3 Weightings

P: Supporting | P: Supporting

= Uncertain Significance

P: Supporting | P: Supporting
P: Strong

= Likely Pathogenic

P: Supporting | P: Supporting
P: Moderate | P: Moderate
P: Strong

= Pathogenic

Functional evidence weighted using the points-based evidence system (**SVC Tavtigian**)

Genetics inMedicine | ORIGINAL RESEARCH ARTICLE

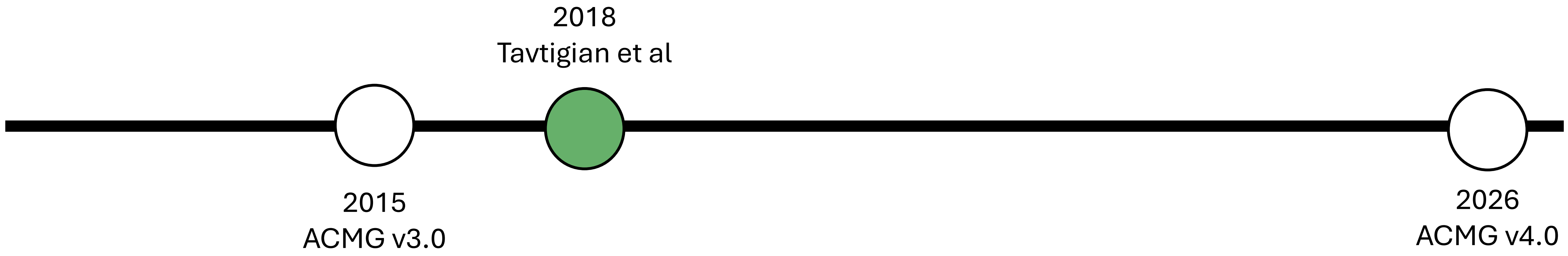
Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework

Sean V. Tavtigian, PhD[1], Marc S. Greenblatt, MD, PhD[2], Steven M. Harrison, PhD[3], Robert L. Nussbaum, MD[4], Snehit A. Prabhu, PhD[5], Kenneth M. Boucher, PhD[6] and Leslie G. Biesecker, MD[7];
on behalf of the ClinGen Sequence Variant Interpretation Working Group (ClinGen SVI)

Key principles:

- ACMG v3.0 classifications align with a naïve Bayesian classification approach

- Increasing strengths scale to the power of 2
    - One piece of Very Strong evidence = 2 pieces of Strong evidence = 4 pieces of Moderate evidence = 8 pieces of Supporting evidence



Tavtigian et al., 2018, Genetics in Medicine

ACMG v3.0 as a Bayesian classifier

Bayes Rule:

$$P(A|B) = \frac{P(B \mid A)P(A)}{P(B)}$$

- P(A): The **prior probability** (the starting assumption)
- P(B): The **marginal probability**
- P(B|A): The **likelihood** of A given B is fixed
- P(A|B): The **posterior probability**

ACMG v3.0 as a Bayesian classifier

Bayes Rule:

$$P(A|B) = \frac{P(B \mid A)P(A)}{P(B)}$$

- P(A): The **prior probability** **of the variant being pathogenic (10%)**
- P(B): The **marginal probability** **being probability of the evidence for pathogenicity occuring**
- P(B|A): The **likelihood** **of the evidence given that the variant is pathogenic**
- P(A|B): The **posterior probability** **of pathogenicity**

ACMG v3.0 as a Bayesian classifier

Simplify to use odds instead…

$$Post\_P = \frac{OddsPath * Prior\_P}{((OddsPath - 1) * Prior\_P + 1}$$

- Prior_P: The **prior probability of the variant being pathogenic (10%)**
- OddsPath: The **likelihood (odds) of the evidence items**
- Post_P: The **posterior probability of pathogenicity**

Tavtigian et al., 2018, Genetics in Medicine

Calculating the OddsPath

$$Post\_P = \frac{OddsPath \ * Prior\_P}{((OddsPath - 1) * Prior\_P + 1}$$

$$OddsPath = \frac{Post\_P \ * (1 - Prior\_P)}{((1 - Post\_P) * Prior\_P}$$

<u>From other sources (prior publications)</u>

Pathogenic = **≥99% confident = 0.99**
Likely Pathogenic = **≥90% confident = 0.90**
Likely Benign **= ≤10% confident = 0.10**
Benign **= ≤0.1% confident = 0.001**

OddsPath **(P)** = 891:1
OddsPath **(LP)** = 81:1
OddsPath **(LB)** = 1 :1
OddsPath **(B)** = 0.009:1

Pathogenic	(i) 1 Very strong (PVS1) *AND* (a) ≥1 Strong (PS1–PS4) *OR* (b) ≥2 Moderate (PM1–PM6) *OR* (c) 1 Moderate (PM1–PM6) and 1 supporting (PP1–PP5) *OR* (d) ≥2 Supporting (PP1–PP5) (ii) ≥2 Strong (PS1–PS4) *OR* (iii) 1 Strong (PS1–PS4) *AND* (a)≥3 Moderate (PM1–PM6) *OR* (b)2 Moderate (PM1–PM6) *AND* ≥2 Supporting (PP1–PP5) *OR* (c)1 Moderate (PM1–PM6) *AND* ≥4 supporting (PP1–PP5)
Likely pathogenic	(i) 1 Very strong (PVS1) *AND* 1 moderate (PM1–PM6) *OR* (ii) 1 Strong (PS1–PS4) *AND* 1–2 moderate (PM1–PM6) *OR* (iii) 1 Strong (PS1–PS4) *AND* ≥2 supporting (PP1–PP5) *OR* (iv) ≥3 Moderate (PM1–PM6) *OR* (v) 2 Moderate (PM1–PM6) *AND* ≥2 supporting (PP1–PP5) *OR* (vi) 1 Moderate (PM1–PM6) *AND* ≥4 supporting (PP1–PP5)

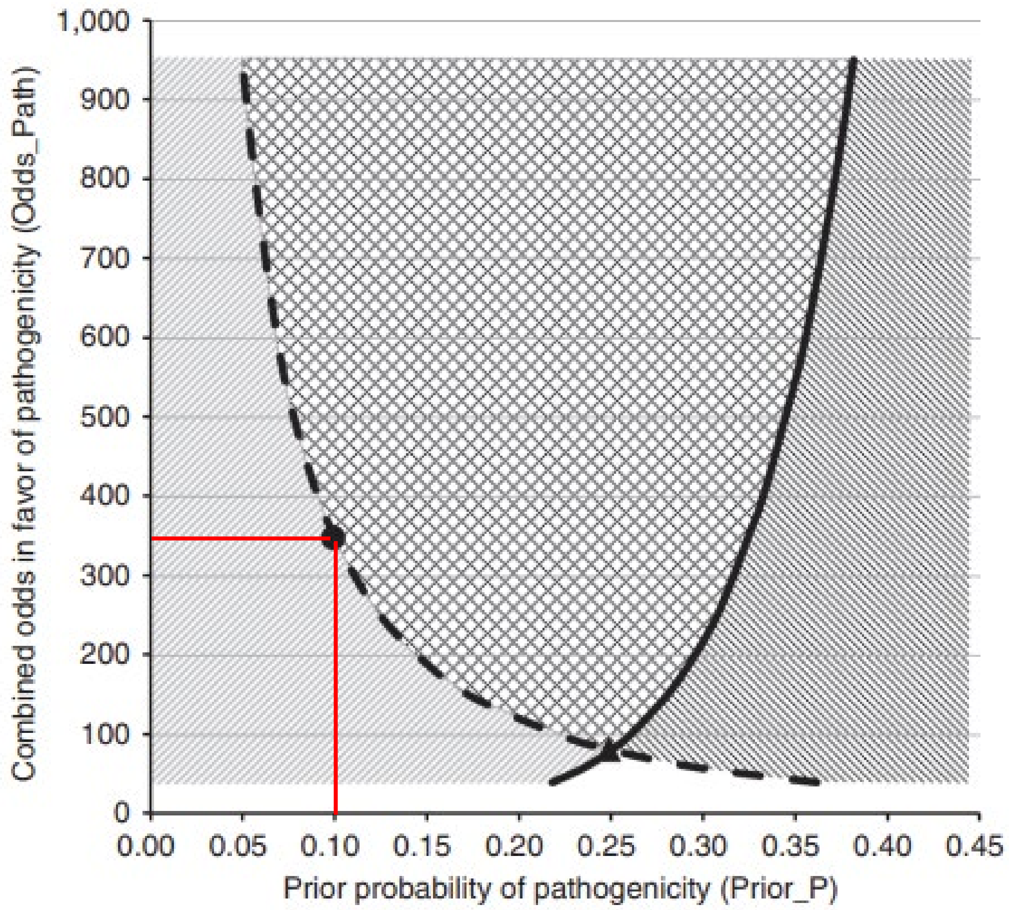
Tavtigian et al., 2018, Genetics in Medicine

Calculating the OddsPath

$$OP = O_{PVSt}^{\left(\frac{N_{PSu}}{X^3}+\frac{N_{PM}}{X^2}+\frac{N_{PSt}}{X}+\frac{N_{PVSt}}{1}\right)}$$

X = Exponent = 2
(1x Very Strong = 2x Strong = 4x Moderate = 8x Supporting pieces of evidence)

O_{PVSt} = Strength of very strong evidence
For prior=0.1 (10%), OPVSt=**350** solves the equation



Tavtigian et al., 2018, Genetics in Medicine

$$Post\_P = \frac{OddsPath * Prior\_P}{((OddsPath - 1) * Prior\_P + 1}$$

	Prior Probability	OddsPath	Posterior Probability	
P: Supporting P: Supporting	**0.1**	$350^{\frac{2Sup}{8}} = 4.3$	**0.32**	**= Uncertain Significance**
P: Supporting P: Supporting P: Strong	**0.1**	$350^{\frac{2Sup}{8}+\frac{1Str}{2}} = 81$	**0.9**	**= Likely Pathogenic**
P: Supporting P: Supporting P: Moderate P: Moderate P: Strong	**0.1**	$350^{\frac{2Sup}{8}+\frac{2Mod}{4}+\frac{1Str}{2}} = 1513.9$	**0.994**	**= Pathogenic**

Functional evidence weighted using the points-based evidence system (**SVC Tavtigian**)

Received: 11 June 2020 | Revised: 18 July 2020 | Accepted: 23 July 2020
DOI: 10.1002/humu.24088

RAPID COMMUNICATION

Human Mutation HGVS HUMAN GENOME VARIATION SOCIETY WILEY

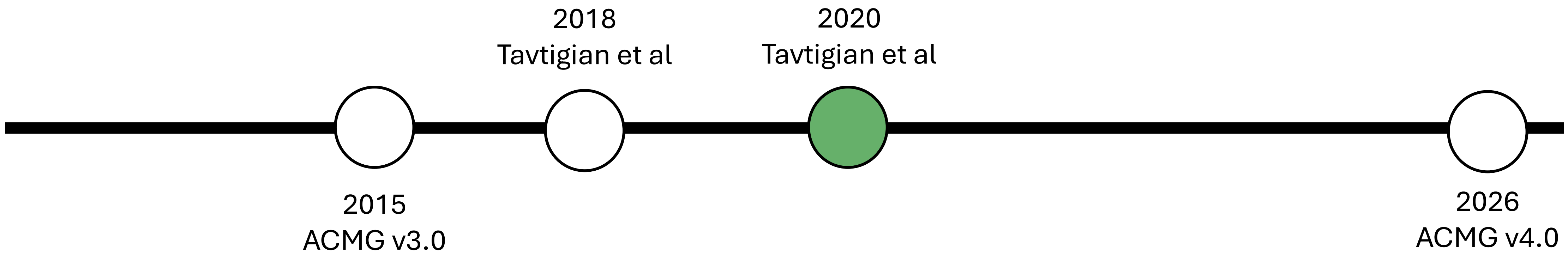
Fitting a naturally scaled point system to the ACMG/AMP variant classification guidelines

Sean V. Tavtigian[1,2] | Steven M. Harrison[3] | Kenneth M. Boucher[2,4] | Leslie G. Biesecker[5]

Key principles:

- Because of the scaled point system, each evidence strength can be given points
  - Very Strong = 8 points
  - Strong = 4 points etc.

- Points can be summed to reach a final classification (at least 6 points overall = Likely Pathogenic)



Tavtigian et al., 2020, Genetics in Medicine

Evidence Strength	Likelihood Odds (OddsPath)	Evidence points ($\log_{2.08}$[OddsPath])
P: Very Strong	350.41	+8
	168.44	+7
	80.98	+6
	38.93	+5
P: Strong	18.71	+4
	9.00	+3
P: Moderate	4.31	+2
P: Supporting	2.08	+1
B: Supporting	0.48	-1
B: Moderate	0.22	-2
B: Strong	0.05	-4
B: Very Strong	0.00	-8

Objective 3: Apply the **ACMG v3.0 SVC Brnich methodology** to assign weighting to a functional assay

Introduction to variant ‘truthsets’

Charlie Rowlands

November 2025

Current practice

General guidelines produced to assist clinical scientists/variant curators in using the data

Functional evidence weighted using the points-based evidence system (**SVC Tavtigian**)

Curation of **variant truthsets**

Clearer methodology (**SVC Brnich**) has been defined for clinical teams and expert groups to assess if a functional assay can be used in classification

Exemplar assay scoring for *BRCA1* assays by CanVIG-UK

Evidence towards pathogenicity   Evidence towards benignity

Gene	Reference	PMID	Evidence Weighting		Please see main CanVIG-UK consensus specification (https://www.cangene-canvaruk.org/canvig-uk-guidance) and gene specific guidance (https://www. results	
			PS3	BS3	Additional CanVIG-UK recommendations	Assay scoring
BRCA1	Bouwman et al, 2020	32546644	Strong	Strong	Bouwman et al, 2020 consists of 3 constituent assays to be scored as 1 assay.	P = the posterior probability for a variant to be deleterious based on its mean-normalised IC50 (olaparib and cisplatin sensitivity assays) or GFP percentage (DR-GFP assay) over 3 biological replicates.
BRCA1	Findlay et al, 2018	30209399	Strong	Strong		HAP1 functional score = cell survival scores, log- normalised across experiments using global medians.
BRCA1	Fernandes et al, 2019	30765603	Supporting	Supporting		Percentage mean of wild type transcriptional activity. Two approaches for transcription data normalisation: 1) vector only expression levels 2) vector and ectopic protein expression levels N.B. 35 variants, all located within the BRCT domain, showed discordant results between the two normalisation methods

A note on splicing assays/MAVEs…

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Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology

Richards et al., Genet Med, 2015

PS3/BS3 (functional evidence) **can** include e.g. minigene assays

ARTICLE

Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup

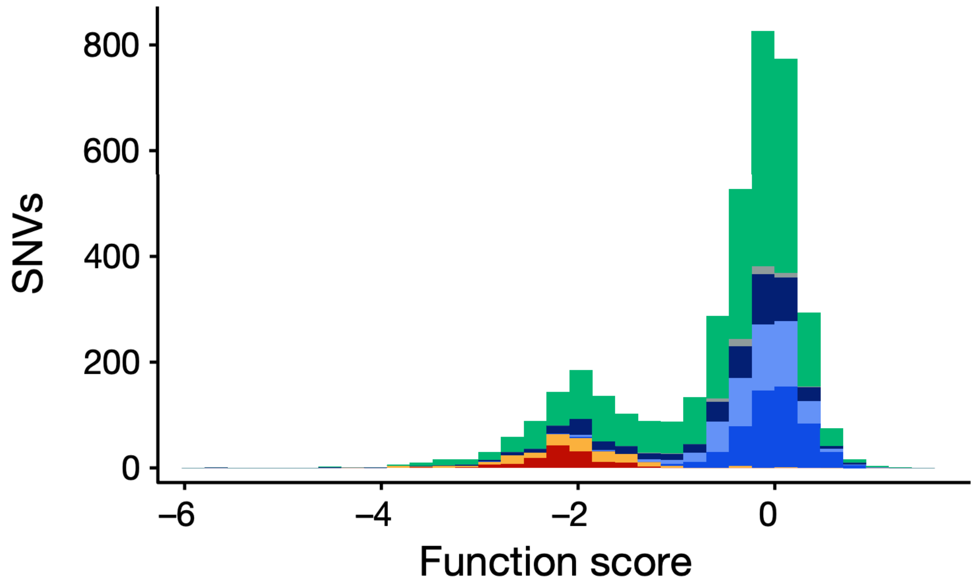
Walker et al., American Journal of Human Genetics, 2023

Splicing assays **removed** from scope of PS3/BS3 (and therefore conventional assays)

Disruption of splicing ≠ pathogenicity!
Lack of impact on splicing ≠ benignity!

However, splicing assays can provide very strong evidence of null variant effect (PVS1)

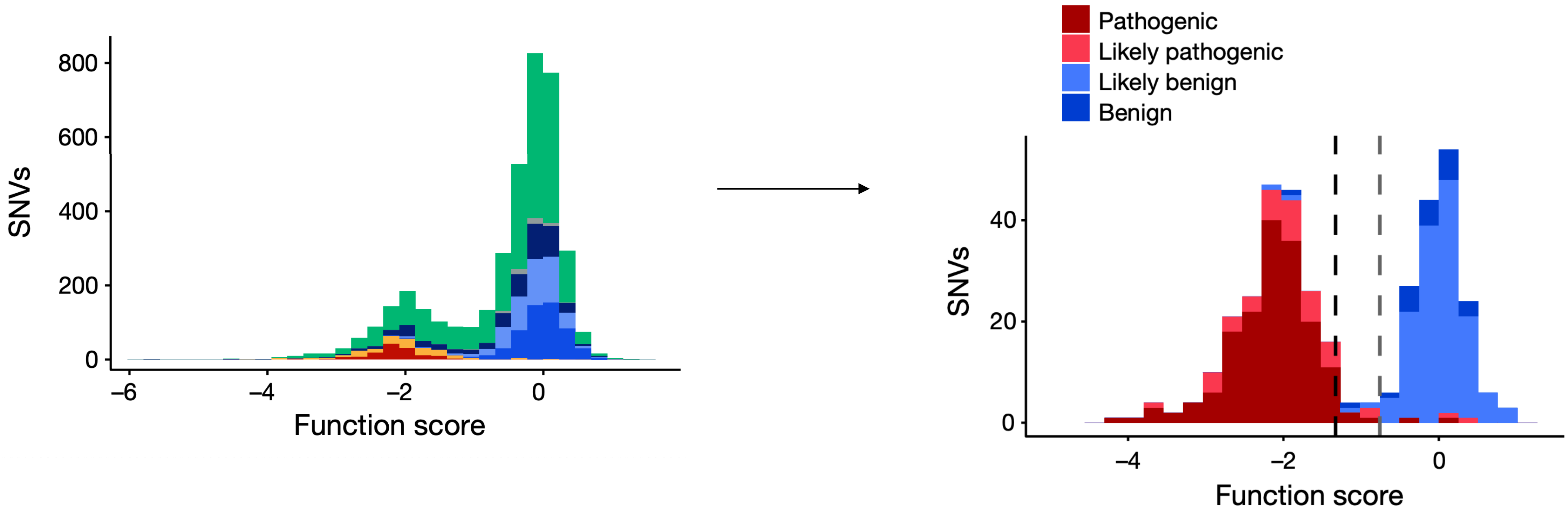
The goal of clinical validation of MAVEs



BRCA1	Findlay et al, 2018	30209399	**Strong**	**Strong**

How well does a MAVE distinguish between pathogenic and benign variants?

The goal of clinical validation of MAVEs



How well does a MAVE distinguish between pathogenic and benign variants?

Dual role of pathogenic and benign truthsets

aka "validation sets", "training sets", "variant sets", "controls"…

Optimal construction of truthsets is critical

Otherwise, it cannot be determined if any observed discrepancies are due to the assay misclassifying or simply sub-optimal truthset construction

Accuracy of assay in discriminating between pathogenic and benign truthset variants

Measure of performance of assay

Number of truthset variants against which assay is validated

Confidence in estimate of performance

PS3 / **BS3**

For any variant with a deleterious assay readout, how much **more** likely is it to be truly pathogenic than truly benign

For any variant with neutral assay readout, how much **more** likely is it to be truly benign than truly pathogenic

The cardinal sin of variant interpretation… double counting

Each item of evidence applied must be independent and applied only once

Explicit

PS1 (strong evidence)

Same amino acid change as a previously established pathogenic variant

PM5 (moderate evidence)

Novel missense change at an amino acid residue where a different pathogenic missense variant has been seen before

Implicit

BA1/BS1/PM2:
population frequency evidence

PP3/BP4:
***in silico* evidence**

Some ***in silico*** tools use **population frequency** to predict pathogenicity/benignity

If using these for variant classification, it is double counting to also use **population frequency** as an independent evidence item!

Selection of variant type ("consequence") for truthsets

Nonsense (for pathogenic)
Synonymous (for benign)

Advantages

- Many (100s-1000s) possible nonsense and synonymous variants for each gene
- Can be modelled in the absence of clinical data
- High probability that they are truly pathogenic and benign

Limitations

- Threshold between pathogenicity and benignity is often calibrated on nonsense/synonymous
- Typically occupy the extremes of assay range → higher prior likelihood of being correctly classified
- Are not typically the most challenging variants in terms of interpretation

Missense (for pathogenic and/or benign)

Advantages

- Exhibit a wider distribution of scores in assays → more rigorous test of nonsense/synonymous-calibrated thresholds
- Most problematic variant type encountered clinically

Limitations

- Many genes have few previously classified pathogenic/benign missense variants

How to construct a truthset

ClinVar
Clinically relevant variation

NM_007294.4(BRCA1):c.5207T>C (p.Val1736Ala)

Cite | Follow | Print | Download

We've updated the full submission template; previous versions are no longer supported. Download the latest version from the ClinVar FTP site and read more about the available templates.

Germline

Reviewed by expert panel | **Pathogenic** | for Breast-ovarian cancer, familial, sus
Classification is based on the expert panel s
Jun 2019 by Evidence-based Network for the In

Somatic

No data submitted for somatic clinical impact

Functional data are available for this variant

ClinVar aggregate review status (star) system

0* = no assertion criteria provided
→ submitter didn't state the classification guidelines they used

1* = criteria provided, single submitter
1* = criteria provided, conflicting classifications

2* = criteria provided, multiple submitters, no conflicts

3* = reviewed by expert panel

4* = practice guidelines

How to construct a truthset

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Clinically relevant variation

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Classification is based on the expert panel s
Jun 2019 by Evidence-based Network for the In

Somatic

No data submitted for somatic clinical impact

Functional data are available for this variant

Types of conflicting classifications (multiple submitters)

"Soft" conflicts

- (Likely) pathogenic and VUS
- (Likely) benign and VUS

"Hard" conflicts

(Likely) pathogenic and (Likely) benign

How to construct a truthset

ClinVar
Clinically relevant variation

CTGATGGTATGGGGCCAAGAGATA
AGGTACGGCTGTCATCACTTAGAC
AGGGCTGGGATAAAAGTCAGGGC
CATGGTGCATCTGACTCCTGAGGA
CAGGTTGGTATCAAGGTTACAAGA
GCACTGACTCTCTCTGCCTATTGG

NM_007294.4(BRCA1):c.5207T>C (p.Val1736Ala)

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Variant Details

Identifiers: NM_007294.4(BRCA1):c.5207T>C (p.Val1736Ala)

Variation ID: 37648 Accession: VCV000037648.49

Type and length: single nucleotide variant, 1 bp

Location: Cytogenetic: 17q21.31 17: 43057122 (GRCh38) [NCBI UCSC] 17: 41209139 (GRCh37) [NCBI UCSC]

Timeline in ClinVar:

	First in ClinVar	Last submission	Last evaluated
Germline	Apr 1, 2014	Jun 29, 2025	Jun 25, 2019

HGVS:

Nucleotide	Protein	Molecular consequence
NM_007294.4:c.5207T>C MANE SELECT	NP_009225.1:p.Val1736Ala	missense
NM_001407571.1:c.4994T>C	NP_001394500.1:p.Val1665Ala	missense
NM_001407581.1:c.5273T>C	NP_001394510.1:p.Val1758Ala	missense

... more HGVS

How to construct a truthset

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Clinically relevant variation

NM_007294.4(BRCA1):c.5207T>C (p.Val1736Ala)

Cite Follow Print Download

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Submissions - Germline

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows
Pathogenic (Jun 25, 2019) C Contributing to aggregate classification	★★★☆ (ENIGMA BRCA1/2 Classification Criteria (2017-06-29))	Breast-ovarian cancer, familial, susceptibility to, 1	Evidence-based Network for the Interpretation of Germline Mutant Alleles (ENIGMA) Accession: SCV001161713.2 First in ClinVar: Feb 20, 2020 Last updated: Jan 07, 2023	

Comment:
IARC class based on posterior probability from multifactorial likelihood analysis, thresholds for class as per Plon et al. 2008 (PMID: 18951446). Class 5 Pathogenic based on posterior probability = 0.999. (less)

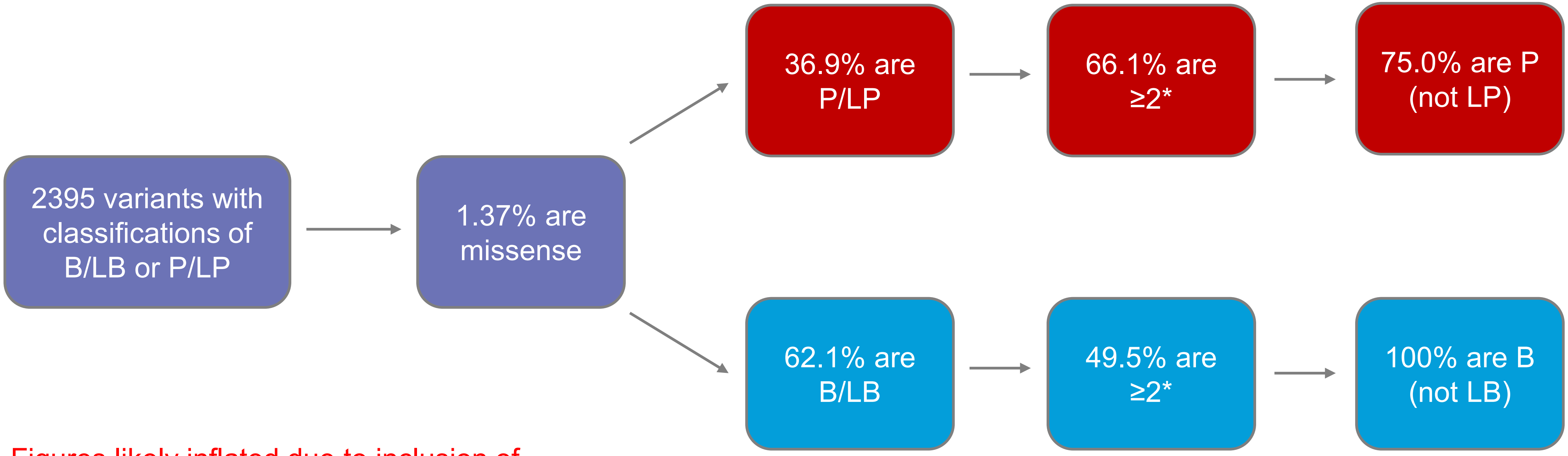
Observation 1

Collection method: curation
Allele origin: germline
Affected status: unknown

Limitations of ClinVar as a truthset source

Few existing high-confidence missense classifications

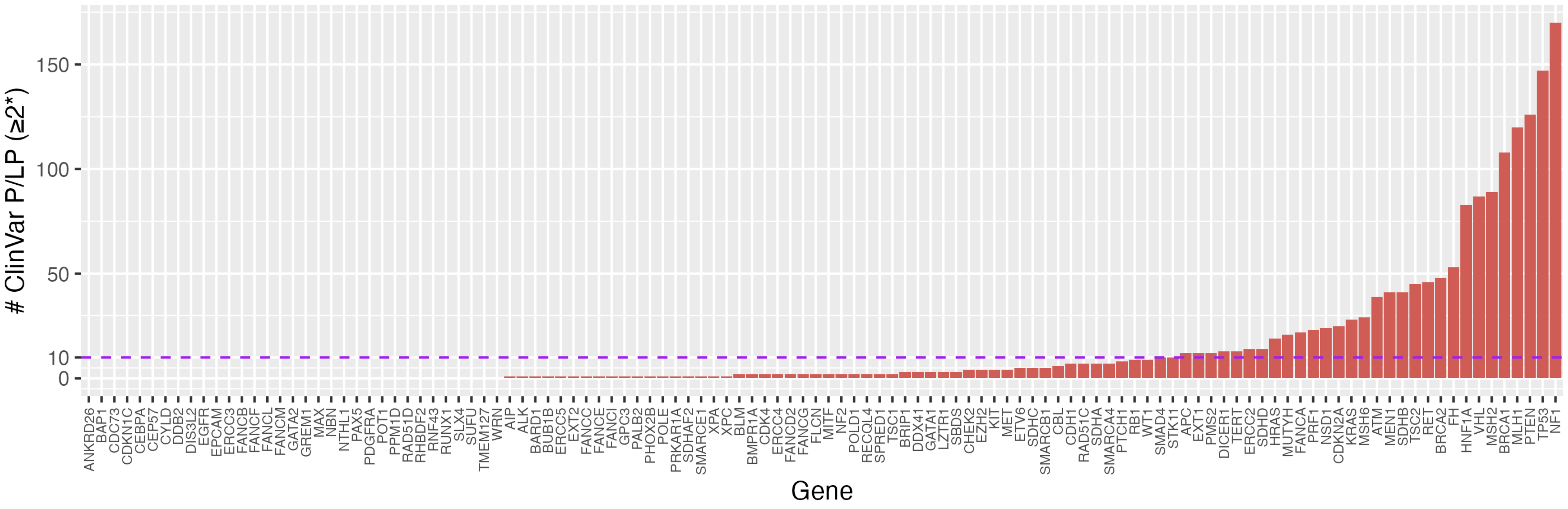
Among 8 hereditary breast and ovarian cancer susceptibility genes, there are a median of…



Figures likely inflated due to inclusion of well-studied genes (*BRCA1* and *BRCA2*)

Limitations of ClinVar as a truthset source

Few existing high-confidence missense classifications



Limitations of ClinVar as a truthset source

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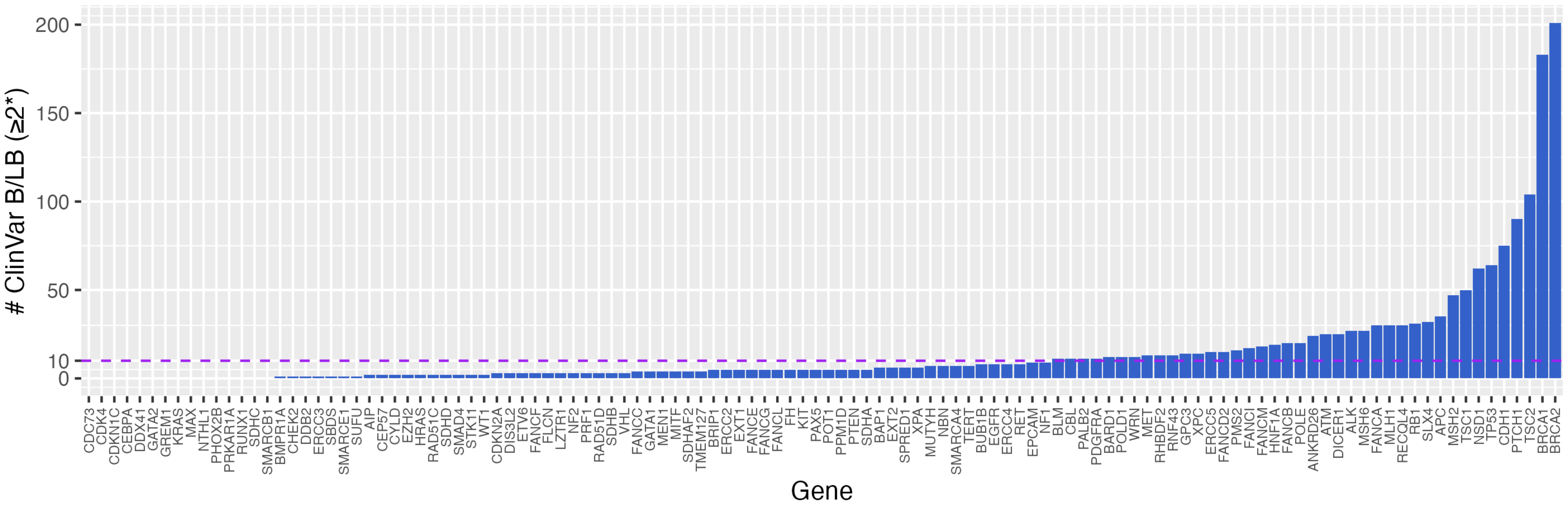
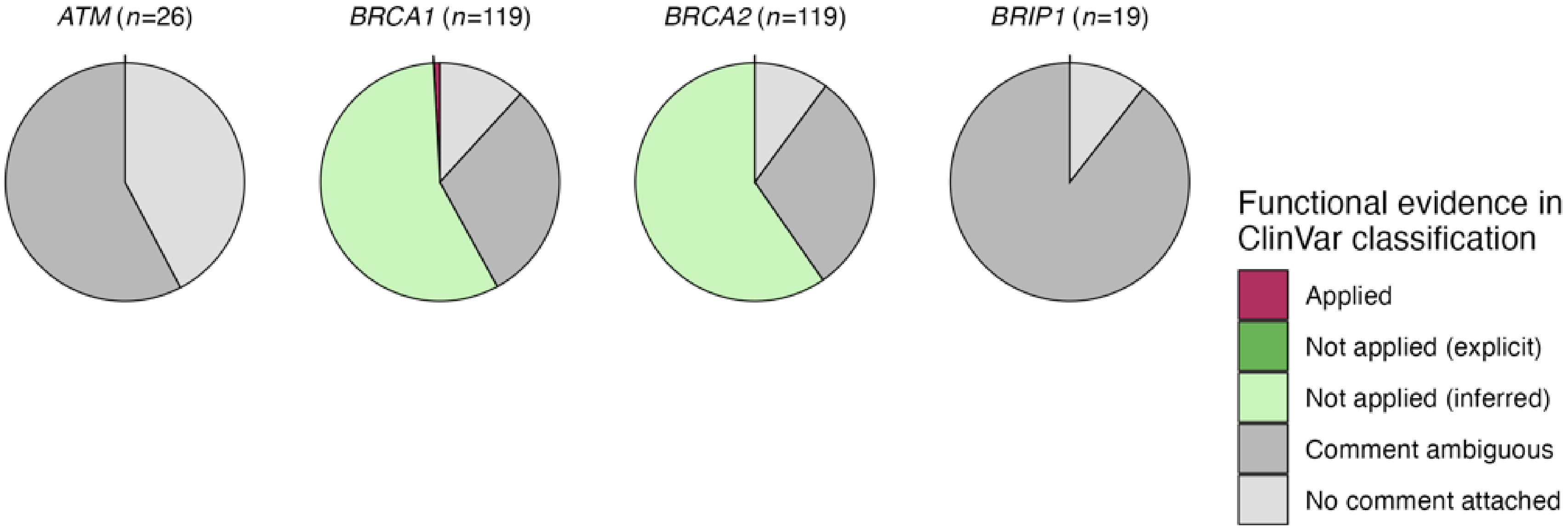


Limitations of ClinVar as a truthset source

Inclusion of functional data in existing classifications (circularity)

Variable quality of comments accompanying submission

1* ClinVar missense variants (single submitter)



Expansion of benign truthsets

Alternative "systematic" approach to benign truthset construction

Population frequency as evidence towards benignity

Population frequency higher than expected for disorder → **BA1**

"Stand-alone" evidence
- Requires no other evidence to reach benignity

Population frequency higher than expected for disorder → **BS1/BS1_sup**

Strong or supporting evidence
- Requires one other evidence item to reach benignity

In silico evidence towards benignity

Computational evidence suggests no impact on gene or gene product → **BP4**

Supporting (or greater) evidence
- Requires one other evidence item to reach benignity

Expansion of benign truthsets

Alternative "systematic" approach to benign truthset construction

***In silico* evidence towards benignity**

BP4

Work conducted by ClinGen Sequence Variant Interpretation (SVI group)

SVI release genome-wide updates to existing evidence codes

Some VCEPs (expert groups) have begun to generate their own gene-specific *in silico* thresholds

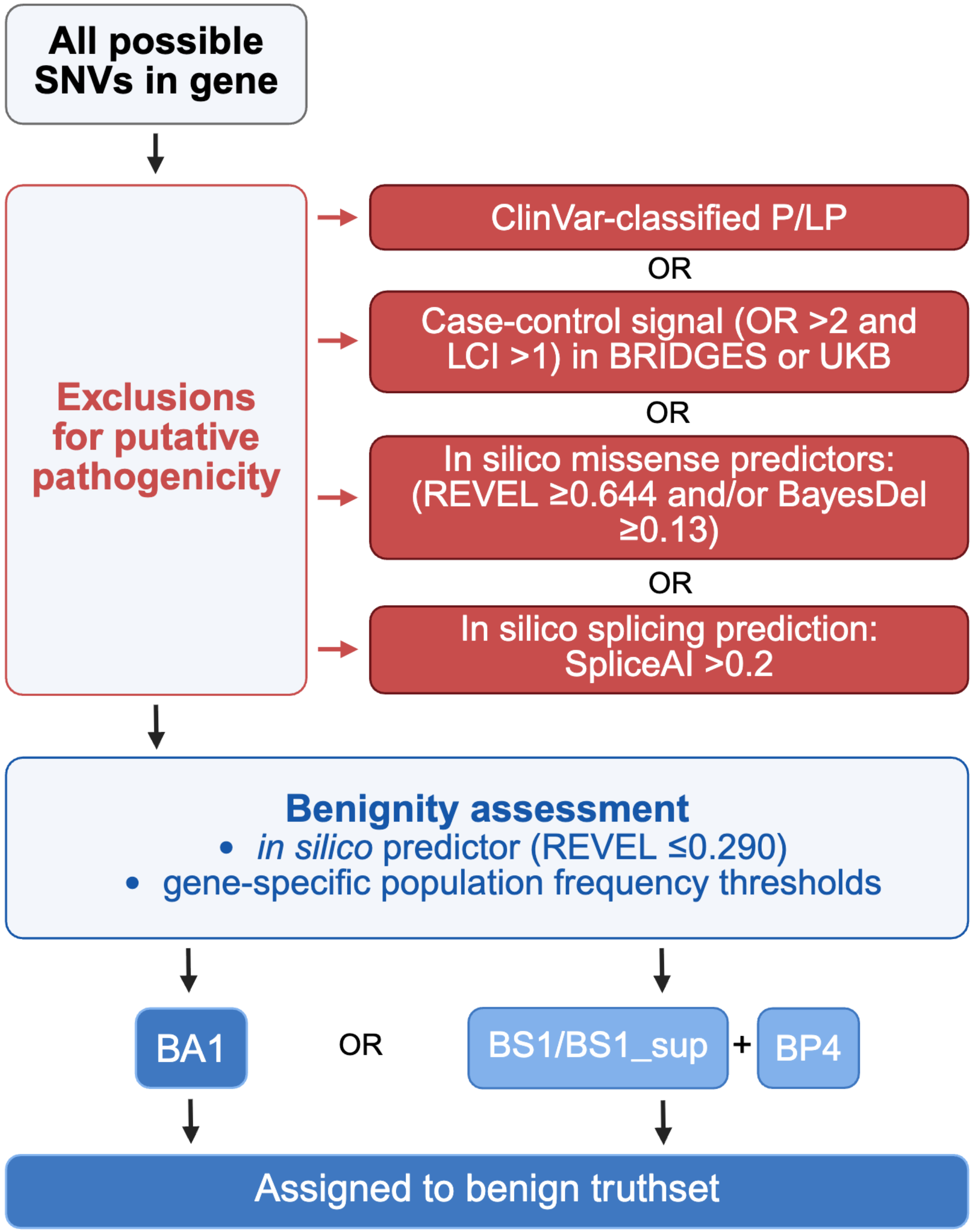
Method	Benign (BP4) Very Strong	Strong	Moderate	Supporting
BayesDel	–	–	≤ −0.36	(−0.36, −0.18]
CADD	–	≤0.15	(0.15, 17.3]	(17.3, 22.7]
EA	–	–	≤0.069	(0.069, 0.262]
FATHMM	–	–	≥4.69	[3.32, 4.69)
GERP++	–	–	≤ −4.54	(−4.54, 2.70]
MPC	–	–	–	–
MutPred2	–	≤0.010	(0.010, 0.197]	(0.197, 0.391]
PhyloP	–	–	≤0.021	(0.021, 1.879]
PolyPhen2	–	–	≤0.009	(0.009, 0.113]
PrimateAI	–	–	≤0.362	(0.362, 0.483]
REVEL	≤0.003	(0.003, 0.016]	(0.016, 0.183]	(0.183, 0.290]
SIFT	–	–	≥0.327	[0.080, 0.327)
VEST4	–	–	≤0.302	(0.302, 0.449]

BayesDel and REVEL most widely used

Pejaver et al., AJHG, 2022

Expansion of benign truthsets

Alternative "systematic" approach to benign truthset construction



BA1/BS1/BS1_sup = allele frequency higher than expected

BP4 = *in silico* scores predict benignity

Rowlands et al., AJHG, 2025

Generation of systematic benign truthsets for three cancer susceptibility genes

BAP1 (Waters et al., 2024)

nature genetics

Article

https://doi.org/10.1038/s41588-024-01799-3

Saturation genome editing of *BAP1* functionally classifies somatic and germline variants

Truthset source	# Path	# Ben	PS3	BS3
ClinVar (≥2*)	0	6	NA	NA
ClinVar (≥1*)	7	6	PS3_mod	BS3_mod
Systematic*	**2,423**	**138**	**PS3_str**	**BS3_vstr**

RAD51C (Olvera-León et al., 2025)

Cell

Article

High-resolution functional mapping of *RAD51C* by saturation genome editing

Authors
Rebeca Olvera-León, Fang Zhang, Victoria Offord, ..., Maria Jasin, Andrew J. Waters, David J. Adams

Truthset source	# Path	# Ben	PS3	BS3
ClinVar (≥2*)	5	2	NA	BS3_sup
ClinVar (≥1*)	8	3	PS3_sup	BS3_mod
Systematic*	**135**	**26**	**PS3_str**	**BS3_str**

POT1 (Obolenski et al., under review)

Truthset source	# Path	# Ben	PS3	BS3
ClinVar (≥2*)	0	5	NA	NA
ClinVar (≥1*)	4	10	PS3_mod	BS3_mod
Systematic*	**234**	**16**	**PS3_mod**	**BS3_str**

All ClinVar counts correct at time of analysis

* Pathogenic truthsets comprising all nonsense

What factors can limit truthset size?

Technical/data availability issues

Few existing high-confidence missense classifications

Especially for ultra-rare or newly disease-associated genes

Disproportionately low deposition of benign variant classifications on ClinVar

Variable quality of classification

Possibility of circularity in genes with existing MAVE datasets

Biological

Gene size (shorter = fewer possible missense variants)
Even more limited when looking at single domain

Inherent tolerance/intolerance to missense variants

- If intolerant, few benign missense (limiting evidence towards pathogenicity) -> issue with *PTEN*
- If tolerant, few pathogenic missense (limiting evidence towards benignity)

N-terminal redundancy (alternative start sites)

Exemplar workflow for truthset construction for a loss-of-function MAVE

Retrieve ClinVar classifications AND comments for gene of interest

Annoyingly bioinformatically challenging!

Great, use them!

Systematically generate them

Enough missense variants?

Great, use them!

Filter for high-confidence ClinVar missense

Enough missense variants?

Supplement with non-splice-impacting synonymous

- Review status >=X* (ideally 2 or 3)
- Evidence that classification was reached without contribution of functional data
  - OR would still reach the same conclusion without the functional data

Supplement with nonsense variants before the last 50 bp of penultimate exon (or pre-defined NMD boundary if known)

Be transparent in detailing how truthsets are constructed (whether clinical or research)

Overview of approaches to truthset construction

BA1/BS1/BS1_sup = allele frequency higher than expected

BP4 = *in silico* scores predict benignity

Truthset variant type (consequence)

Pathogenic

?Ideally

- Missense
- Nonsense

*before last 50 bp of penultimate exon

Benign

?Ideally

- Missense
- Synonymous

*with no evidence of splicing impact

Truthset variant source

ClinVar

Review status

≥3*	≥2*	≥1*
Reviewed by expert panel	Multiple submitters, no conflicts	Single submitter

Conflicting classifications

Inclusion of soft conflicts

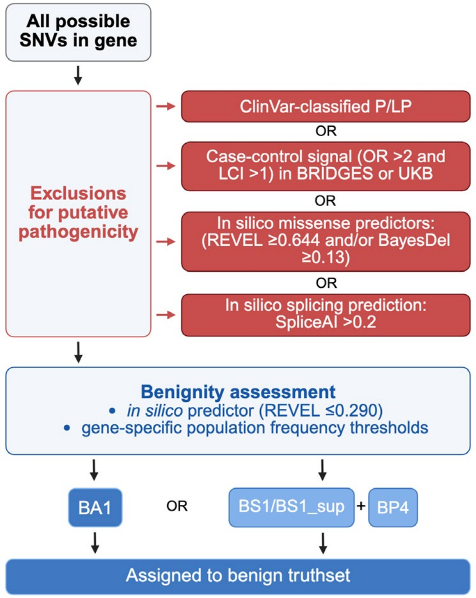
or exclusion

(Likely) pathogenic and **VUS**, or
(Likely) benign and **VUS**

Final classification strength

Pathogenic + Benign OR **(Likely) pathogenic + (Likely) benign**

Systematically generated



Practice Session 1: Defining Variant Truthsets

November 2025

Please open Part 1 of the exercise

Practice Session Overview (Part 1)

We have data for two large-scale MAVE assays (one for *BRCA1*, and one for *VHL*). You will each be assigned a gene to start with.

Your task:

Starting with your assigned gene (either *VHL* or *BRCA1*), perform the following filters for each respective tab.

For each tab, count the number of variants in your Pathogenic and Benign truthsets (as defined on your session handout)

Practice Session Overview (Part 1)

Record on each tab the number of variants in your Pathogenic and Benign truthsets (as defined below, and with further detail in your session handout):

1. ClinVar ≥ 1*, all variant types, pathogenic/likely pathogenic in ClinVar and benign/likely benign in ClinVar
2. ClinVar ≥ 2*, all variant types, pathogenic/likely pathogenic in ClinVar and benign/likely benign in ClinVar
3. ClinVar ≥ 1* plus conflicting classifications, all variant types, any classification containing a pathogenic/likely pathogenic or benign/likely benign classification in ClinVar
4. ClinVar ≥ 1*, missense variants, pathogenic/likely pathogenic in ClinVar and benign/likely benign in ClinVar
5. ClinVar ≥ 1*, missense variants, pathogenic/likely pathogenic in ClinVar, and systematic benign variants.
6. All nonsense and synonymous variants

Objective 3: Apply the **ACMG v3.0 SVC Brnich methodology** to assign weighting to a functional assay

The Brnich et al Methodology

Charlie Rowlands

November 2025

Current practice

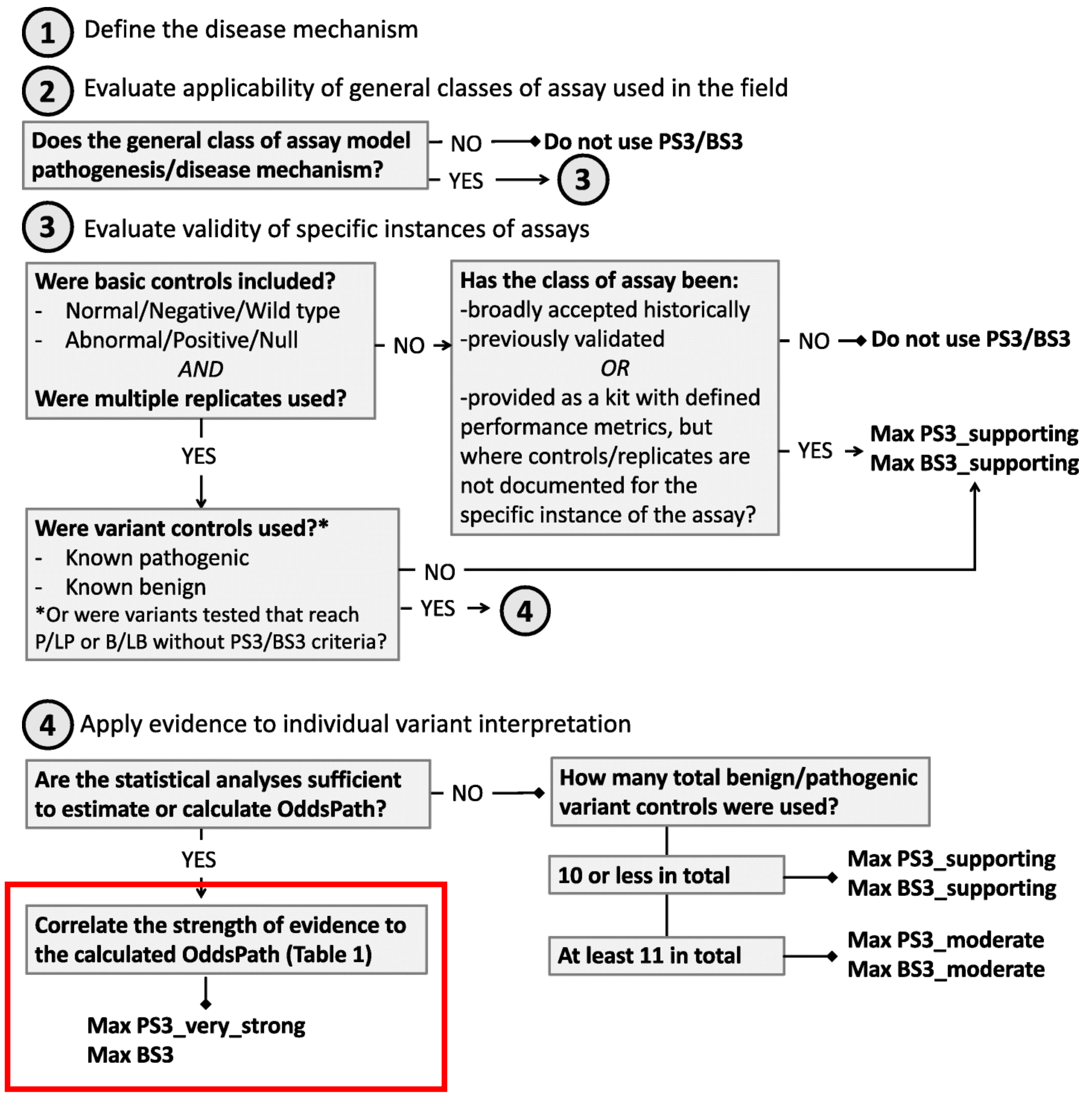
General guidelines produced to assist clinical scientists/variant curators in using the data

Functional evidence weighted using the points-based evidence system (**SVC Tavtigian**)

Curation of **variant truthsets**

Clearer methodology (**SVC Brnich**) has been defined for clinical teams and expert groups to assess if a functional assay can be used in classification

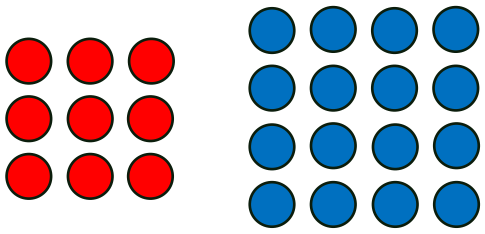
Brnich et al. framework for assay strength quantification



Worked example of Brnich framework

1. Calculate prior probability of **pathogenicity** based on **relative number of total truthset variants**

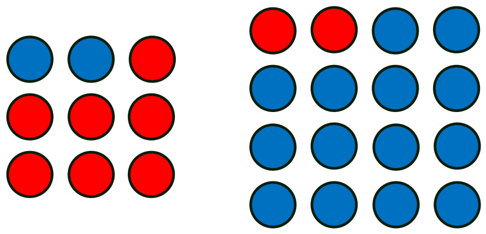
Path. Ben.



$$Prior_{Path} = \frac{9}{9+16} = 0.36$$

2. Calculate posterior probability of **pathogenicity** based on **relative numbers in each class of assay output**

Deleterious Neutral



$$Post_{Path(Del)} = \frac{7}{9+1} = 0.70$$

$$Post_{Path(Neu)} = \frac{2+1}{16+1} = 0.18$$

3. Calculate **Odds$_{Path}$** for each of deleterious and neutral readouts

$$Post\_P = \frac{OddsPath * Prior\_P}{((OddsPath - 1) * Prior\_P + 1}$$

$$\boldsymbol{Odds_{Path}} = \frac{\boldsymbol{Post_{Path} * (1 - Prior_{Path})}}{\boldsymbol{(1 - Post_{Path}) * Prior_{Path}}}$$

$$\boldsymbol{Odds_{Path(Del)} = 4.15} \qquad \boldsymbol{Odds_{Path(Neu)} = 0.38}$$

4. Calculate **Odds$_{Path}$** to **PS3**/**BS3** **evidence strengths** or **EPs**

Evidence Strength	Likelihood Odds (OddsPath)	Evidence points ($\log_{2.08}$[OddsPath])
P: Very Strong	350.41	+8
	168.44	+7
	80.98	+6
	38.93	+5
P: Strong	18.71	+4
	9.00	+3
P: Moderate	4.31	+2
P: Supporting	2.08	+1
B: Supporting	0.48	-1
B: Moderate	0.22	-2
B: Strong	0.05	-4
B: Very Strong	0.00	-8

Converting OddsPath to an applicable evidence strength

Evidence Strength	Likelihood Odds (OddsPath)	Evidence points ($\log_{2.08}$[OddsPath])
P: Very Strong	350.41	+8
	168.44	+7
	80.98	+6
	38.93	+5
P: Strong	18.71	+4
	9.00	+3
P: Moderate	4.31	+2
P: Supporting	2.08	+1
B: Supporting	0.48	-1
B: Moderate	0.22	-2
B: Strong	0.05	-4
B: Very Strong	0.00	-8

$$Odds_{Path(Del)} = 4.15$$

$$EPs_{PS3} = \mathbf{1.94}$$

$$Strength_{PS3} = Supporting$$

$$Odds_{Path(Neu)} = 0.38$$

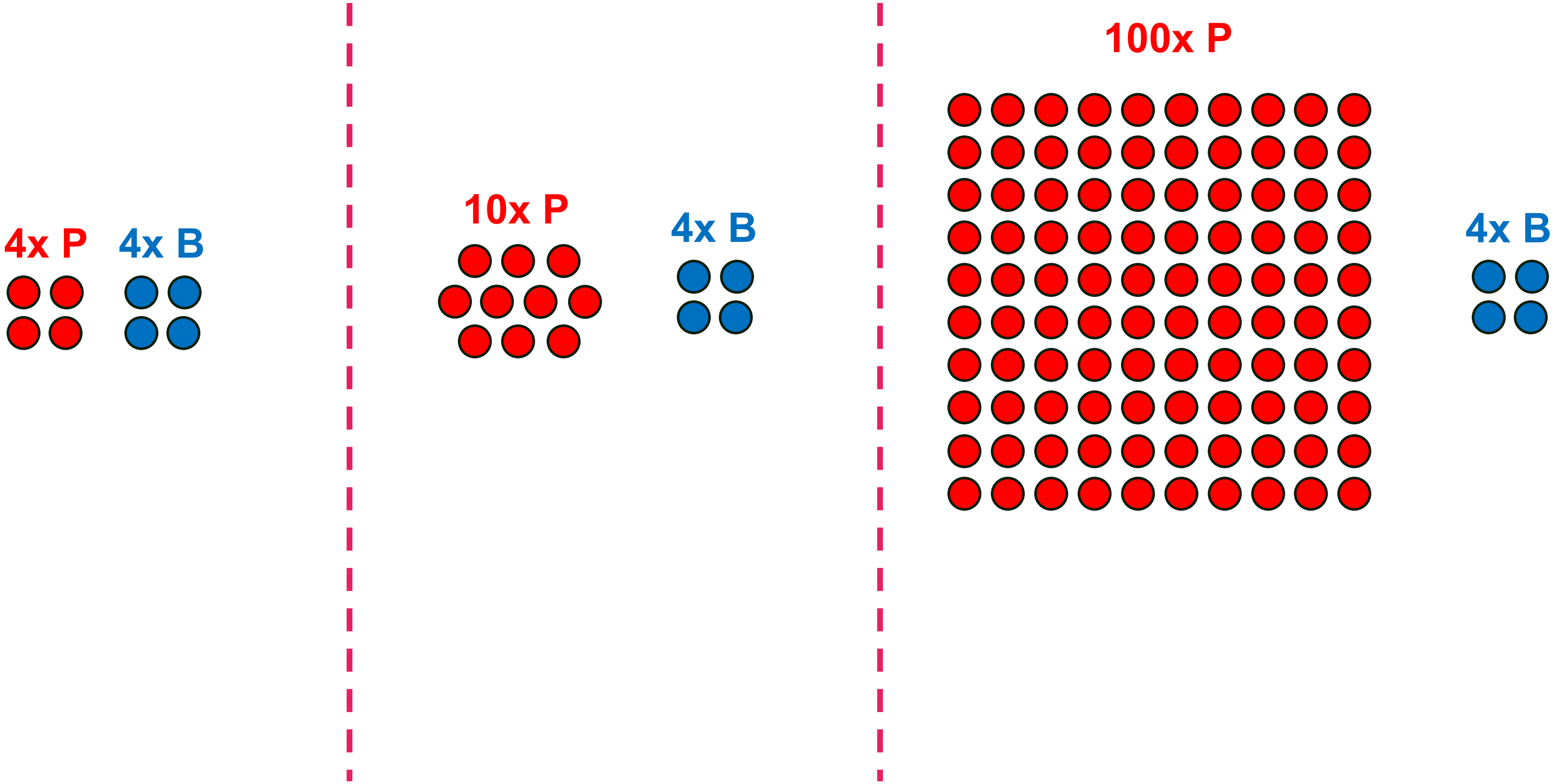
$$EPs_{BS3} = \mathbf{-1.32}$$

$$Strength_{BS3} = Supporting$$

Pathogenic truthset size drives applicable BS3 strength

PS3 = evidence towards **pathogenicity**

BS3 = evidence towards **benignity**



PS3 EPs (strength)

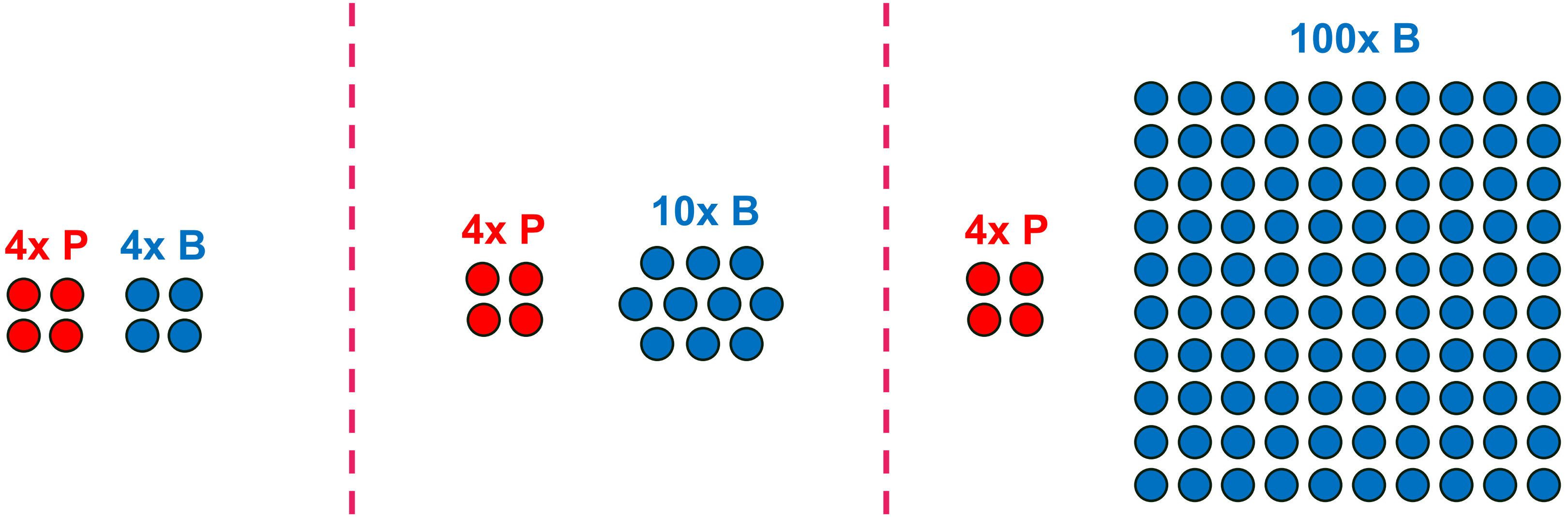
BS3 EPs (strength)

**Assuming perfect discrimination between pathogenic and benign truthset variants*

Benign truthset size drives applicable BS3 strength

PS3 = evidence towards **pathogenicity**

BS3 = evidence towards **benignity**



PS3 EPs (strength)	1.89 (**supporting**)
BS3 EPs (strength)	1.89 (**supporting**)

**Assuming perfect discrimination between pathogenic and benign truthset variants*

Effect of misclassifications on PS3/BS3 strength

# truthset variants		# variants with deleterious readout		# variants with neutral readout			
Pathogenic	Benign	Pathogenic	Benign	Pathogenic	Benign	PS3 EPs	BS3 EPs
9	9	9	0	0	9	3.00	3.00
18	9	18	0	0	9	3.00	3.94
36	9	36	0	0	9	3.00	4.89
37	9	36	0	1	9	2.96	3.98
74	9	73	0	1	9	2.98	4.93
74	74	73	0	1	74	5.85	4.93

Double path. variants = +1 BS3 point

Double path. variants = +1 BS3 point

Add one misclassified pathogenic = -1 BS3 point

Add one misclassified pathogenic = -1 BS3 point

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Practice Session 2: Assessing evidence weighting

November 2025

Please open Part 2 of the exercise

Practice Session Overview (Part 2)

In the first exercise, you defined truthsets of variants for two large-scale assays (*BRCA1* and *VHL*).

Your task: Using your six pre-filtered truthsets from Part 1, assess the evidence weighting which should be applied for each assay

(for each assay we are assuming both the disease mechanism and assay class are applicable, and both assays are robust and reproducible)

Pre-filtered google sheet from Part 1: https://docs.google.com/spreadsheets/d/1W9vHBKsAyP6GXRNqIGUFWBBbYeCM6zHVdi21BzZMHo4/edit?gid=0#gid=0&fvid=1827307889

Link to Part 2: https://docs.google.com/spreadsheets/d/1aUJTWwB7uETnkn-qMpzB0NIrqvfOt2mzEiWg-GeGRLY/edit?gid=0#gid=0

Practice Session Example

Truthset Name	Truthset Content	# Classified			Proportion pathogenic (Prior, P1)	Assay result						"+1 proportions" (Posterior, P2)		OddsPath=[P2 x (1- P1)] / [(1- P2) x P1]	
						Pathogenic truthset			Benign truthset			Pathogenic	Benign	Pathogenic	Benign
		Total	Pathogenic truthset	Benign truthset		Functionally abnormal	Intermediate	Functionally normal	Functionally abnormal	Intermediate	Functionally normal				
EXAMPLE	EXAMPLE	17	10	7	0.59	10	0	0	1	0	6	0.833	0.143	3.475	0.116

For an assay with:

- 10 variants in our **pathogenic** truthset
- 7 variants in our **benign** truthset

Of these variants:

- All 10 of the Pathogenic truthset variants are 'functionally abnormal'
- 6 of the benign truthset variants are 'functionally normal' and 1 is 'functionally abnormal'

Spreadsheet will calculate OddsPath automatically – you can use this to determine the evidence strength using the provided conversion table

Practice Session Feedback (*BRCA1*)

Truthset Name	Truthset Content	# Classified			Proportion pathogenic (Prior, P_1)	Assay result						"+1 proportions" (Posterior, P_2)		OddsPath=$[P_2 \times (1- P_1)] / [(1- P_2) \times P_1]$	
						Pathogenic Truthset			Benign Truthset			Pathogenic	Benign	Pathogenic	Benign
		Total	Pathogenic Truthset	Benign Truthset		Functionally abnormal	Intermediate	Functionally normal	Functionally abnormal	Intermediate	Functionally normal				
Truthset 1	BRCA1 (ClinVar ≥1*)	1004	438	566	0.44	422	12	4	7	11	548	0.98	0.01	68.2	0.012
Truthset 2	BRCA1 (ClinVar ≥2*)	608	331	277	0.54	318	10	3	4	9	264	0.98	0.01	53.2	0.013
Truthset 3	BRCA1 (ClinVar ≥1* plus conflicts)	1378	516	862	0.37	490	17	9	8	16	838	0.98	0.01	91.0	0.020
Truthset 4	BRCA1 (ClinVar ≥1*, missense only)	284	187	97	0.66	181	4	2	1	1	95	0.99	0.03	46.9	0.016
Truthset 5	BRCA1 (ClinVar ≥1* and systematic benign, missense only)	188	187	1	0.99	181	4	2	0	0	1	0.99	0.75	1.0	0.016
Truthset 6	BRCA1 (Nonsense + synonymous)	682	138	579	0.20	136	2	0	11	12	556	0.92	0.00	47.6	0.008

Practice Session Feedback (*VHL*)

Truthset Name	Truthset Content	# Classified: Total	# Classified: Pathogenic Truthset	# Classified: Benign Truthset	Proportion pathogenic (Prior, P_1)	Assay result: Pathogenic Truthset: Functionally abnormal	Assay result: Pathogenic Truthset: Intermediate	Assay result: Pathogenic Truthset: Functionally normal	Assay result: Benign Truthset: Functionally abnormal	Assay result: Benign Truthset: Intermediate	Assay result: Benign Truthset: Functionally normal	"+1 proportions" (Posterior, P_2): Pathogenic	"+1 proportions" (Posterior, P_2): Benign	OddsPath=$[P_2 \times (1-P_1)] / [(1-P_2) \times P_1]$: Pathogenic	OddsPath=$[P_2 \times (1-P_1)] / [(1-P_2) \times P_1]$: Benign
Truthset 1	VHL (ClinVar ≥1*)	510	225	285	0.44	173	15	37	8	12	265	0.95	0.13	24.3	0.182
Truthset 2	VHL (ClinVar ≥2*)	287	134	153	0.47	112	10	12	2	6	145	0.97	0.08	42.6	0.102
Truthset 3	VHL (ClinVar ≥1* plus conflicts)	611	268	343	0.44	190	21	57	9	15	319	0.95	0.15	24.3	0.233
Truthset 4	VHL (ClinVar ≥1*, missense only)	164	153	11	0.93	121	11	21	0	1	10	0.99	0.69	8.7	0.158
Truthset 5	VHL (ClinVar ≥1* and systematic benign, missense only)	313	153	160	0.49	121	11	21	5	9	146	0.95	0.13	21.1	0.158
Truthset 6	VHL (Nonsense + synonymous)	434	60	386	0.14	43	2	15	13	10	363	0.75	0.04	19.8	0.280

Practice Session Feedback

Discussion

Is this the same as the evidence weighting you assigned? Why/Why not?

Of the approaches used to calculate (or construct) evidence weighting, which do you think is the most appropriate?

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Objective 4: Learn about new and upcoming methodologies used for **variant-level weighting**

Link to recording:
https://drive.google.com/file/d/1PJ6GjFdmC7PzAeyJdwTx30IsNNeDl6y4/view?usp=drive_link

Variant-level evidence weighting

Sophie Allen (Sophie.Allen@icr.ac.uk)

November 2025

General Background

Brnich-style OddsPath approach

Using author-defined thresholds

1. Standard **dichotomous** Brnich : 2 (+1) zones (non-functional, (intermediate), neutral
2. Single evidence strength for variants towards/against pathogenicity

An alternative approach: Variant-level scoring

Disregard author-defined thresholds. Generate new thresholds based on data

1. Output on a **quantitative** scale (likelihood-based outputs)
    1. This means obtaining **variant-level evidence weighting**
2. Can also output the 9 evidence classes (per ACMG v3.0)

General Function

Required data: Assay function score, labelled variant truthset (class 1: pathogenic, class 2: benign)

Step 1: Model trained on reference variant set to identify a discriminant threshold between two classes

- **For our purposes: using two classes (P/LP and B/LB) from a ClinVar reference set.**

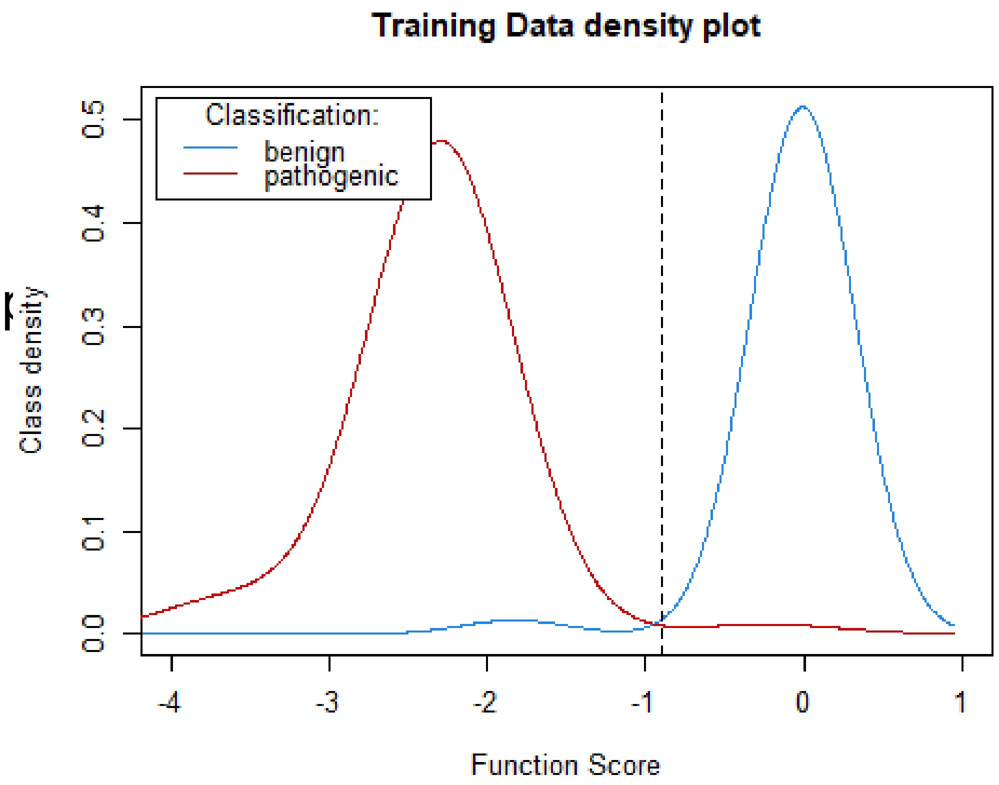
Step 2: Trained model used on remaining unlabelled data to identify which class the data belongs to

- Data = assay score
- The further from the threshold assay score is = stronger evidence towards pathogenicity or benignity

Gaussian Mixture Model (GMM)

In practice
Application with a skew-normal distribution
Zeiberg et al., 2025 (preprint)
https://doi.org/10.1101/2025.04.29.651326

Assumes that both class distributions follow a **Gaussian curve:**

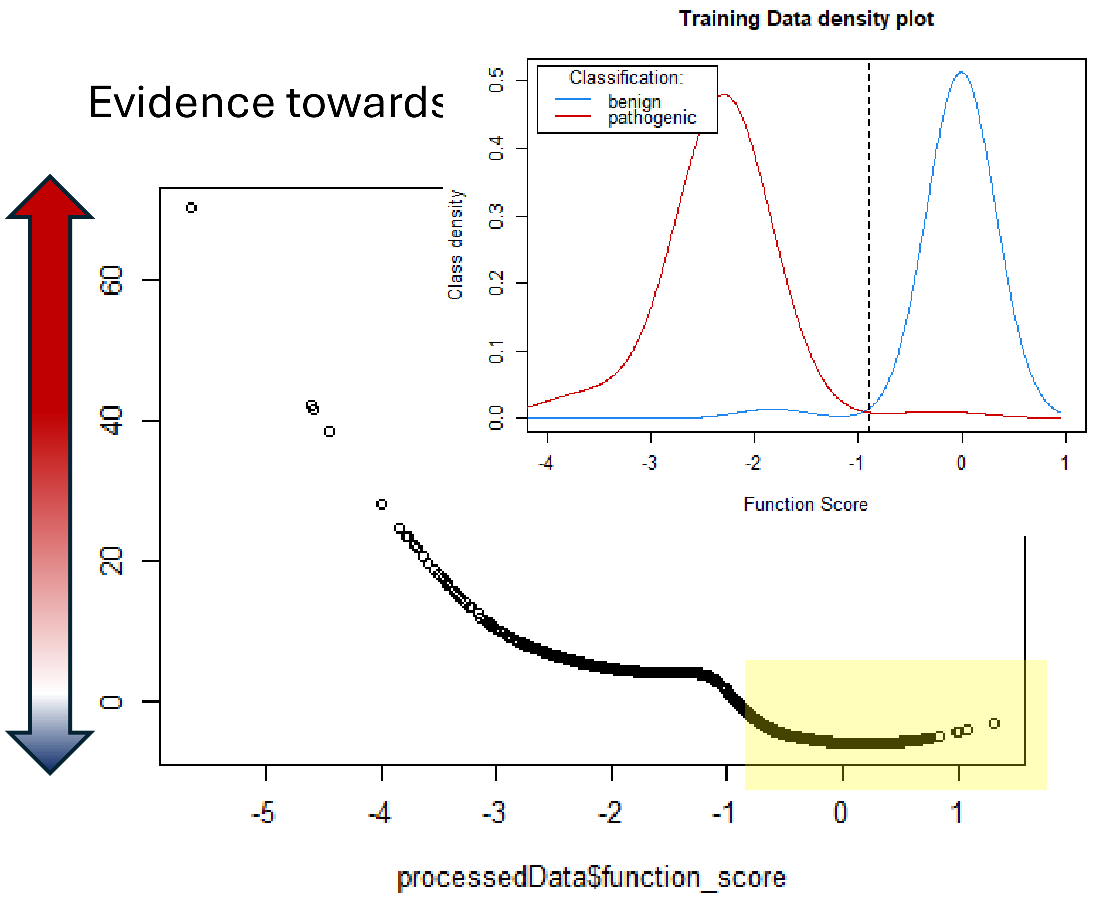


Considerations

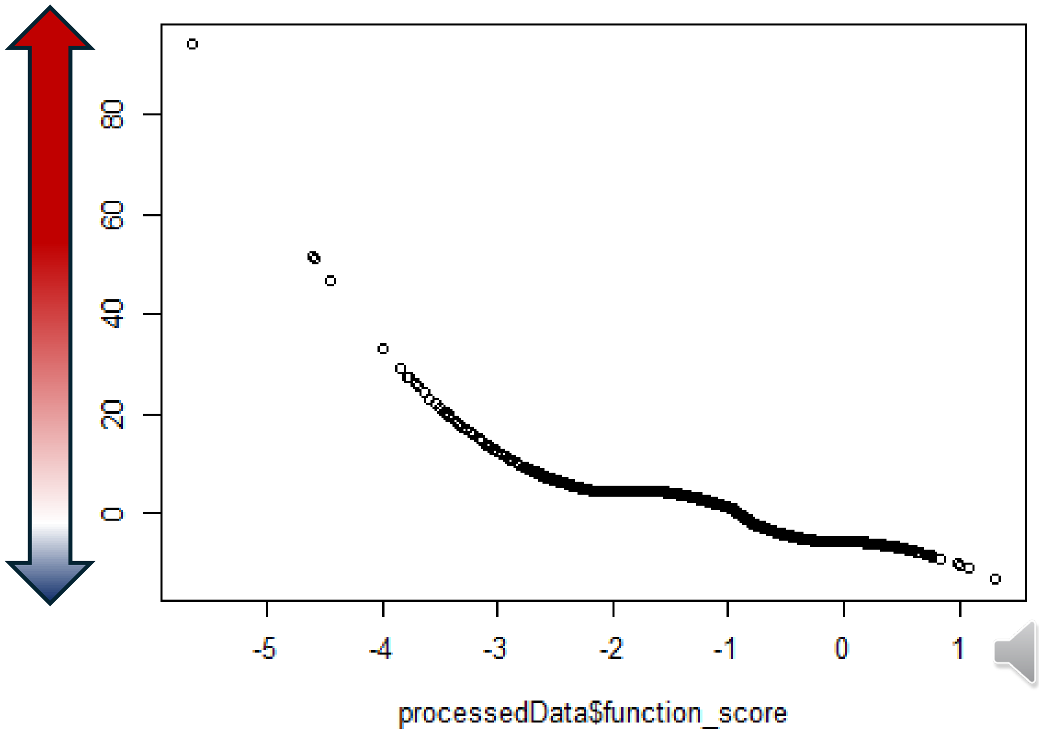
- Model assumes equal priors when training
- Distribution of the two classes assumed to be Gaussian [can adjust with skew-normal distributions]
- Bootstrapping
- Monotonic constraints are needed to avoid extreme ends of model tails impacting overall

Adjusting for monotonic constraints

Evidence towards



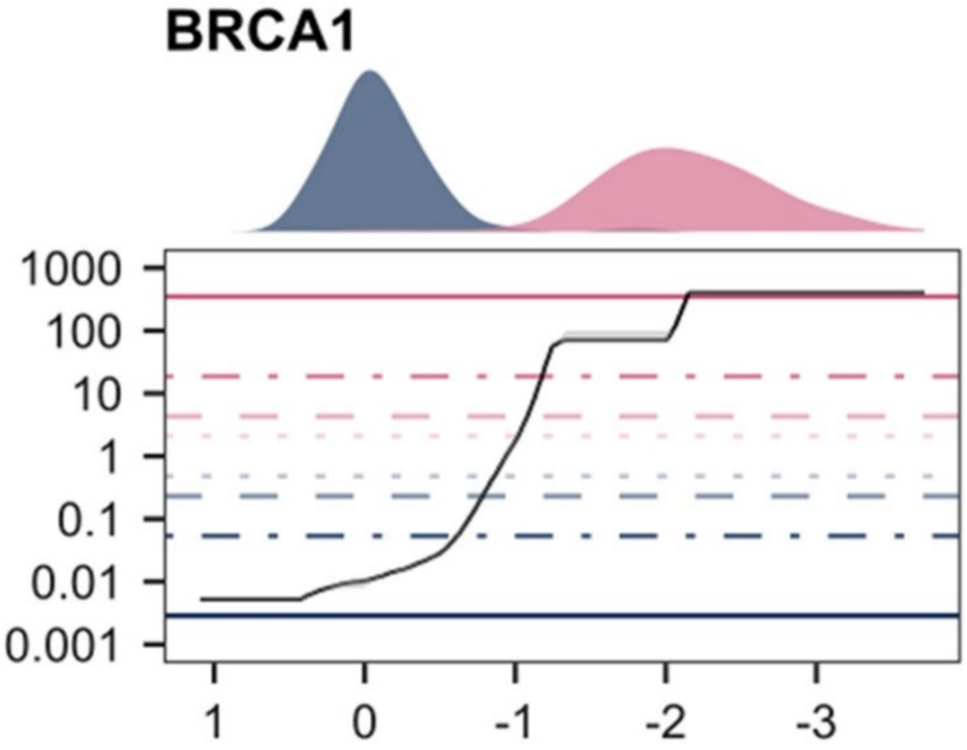
Evidence towards pathogenicity: constraints added



Kernel Density Estimation (KDE)

In practice
Gebbia et al., 2024. AJHG (CHEK2)
https://doi.org/10.1016/j.ajhg.2024.10.013
van Loggerenberg et al., 2023. AJHG (HMBS)
https://doi.org/10.1016/j.ajhg.2023.08.012

Plots **probability density function** for each class:



Badonyi and Marsh, 2025. Bioinformatics.

Considerations

- Bandwidth may need tuning
- Bootstrapping
- Monotonic constraints are needed to avoid extreme ends of model tails impacting overall

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Example

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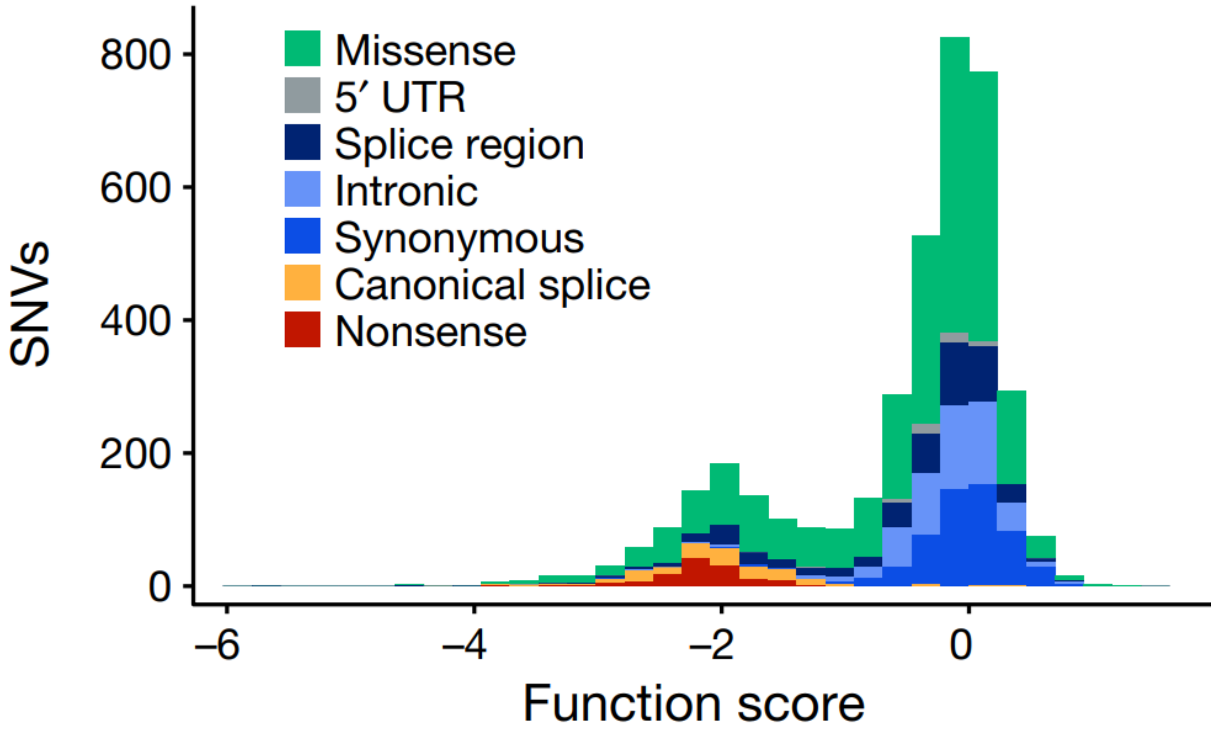
Example

2018 ClinVar dataset (all stars), Findlay et al 2018 BRCA1 dataset

Identified 477 variants in the reference variant set (209 B/LB and 268 P/LP)



Using the Brnich et al methodology:

- 253/268 P/LP variants reported as LOF by the assay
- 199/209 B/LB variants reported as Functional by the assay

OddsPath, pathogenicity = 197.3 (PS3_strong)
OddsPath, benignity = 0.004 (BS3_strong)

Findlay et al., 2018. Nature.

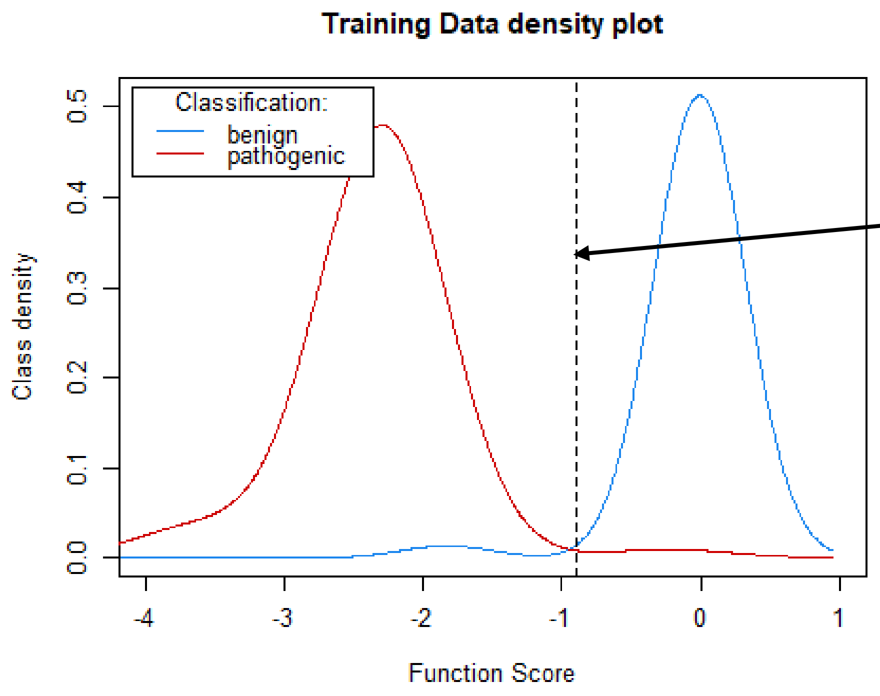
Example

2018 ClinVar dataset (all stars), Findlay et al 2018 BRCA1 dataset

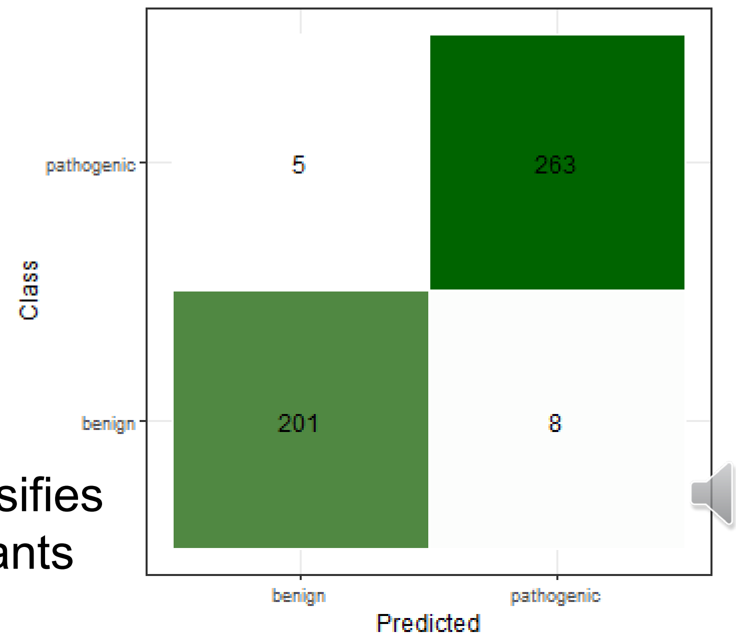
Identified 477 variants in the reference variant set (209 B/LB and 268 P/LP)

- Remove variants in the reference set from the main assay set (leaving 3416 variants for testing)

- Train the model using the reference set, model places discriminant threshold between P/LP and B/LB function scores



GMM places the threshold at a function score of -0.90



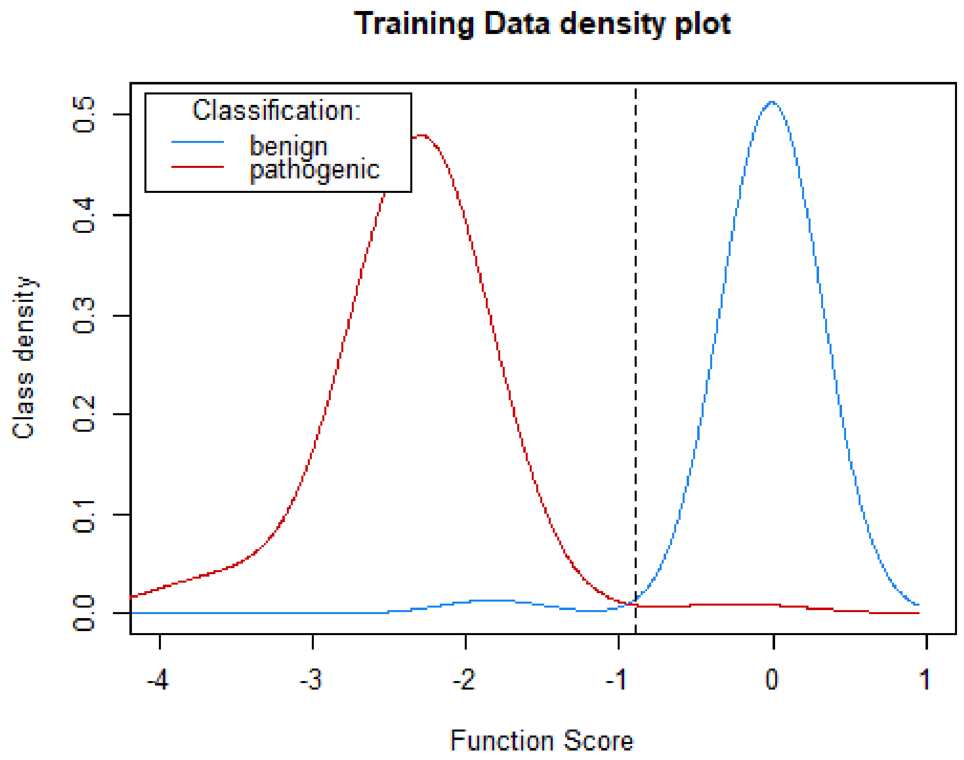
Correctly classifies 97.2% of variants

Example

2018 ClinVar dataset (all stars), Findlay et al 2018 BRCA1 dataset

Comparison of the density estimate for each of the pathogenic and benign distributions (density estimates being roughly equivalent to likelihood odds)

Pathogenic density / Benign density = LR (evidence)

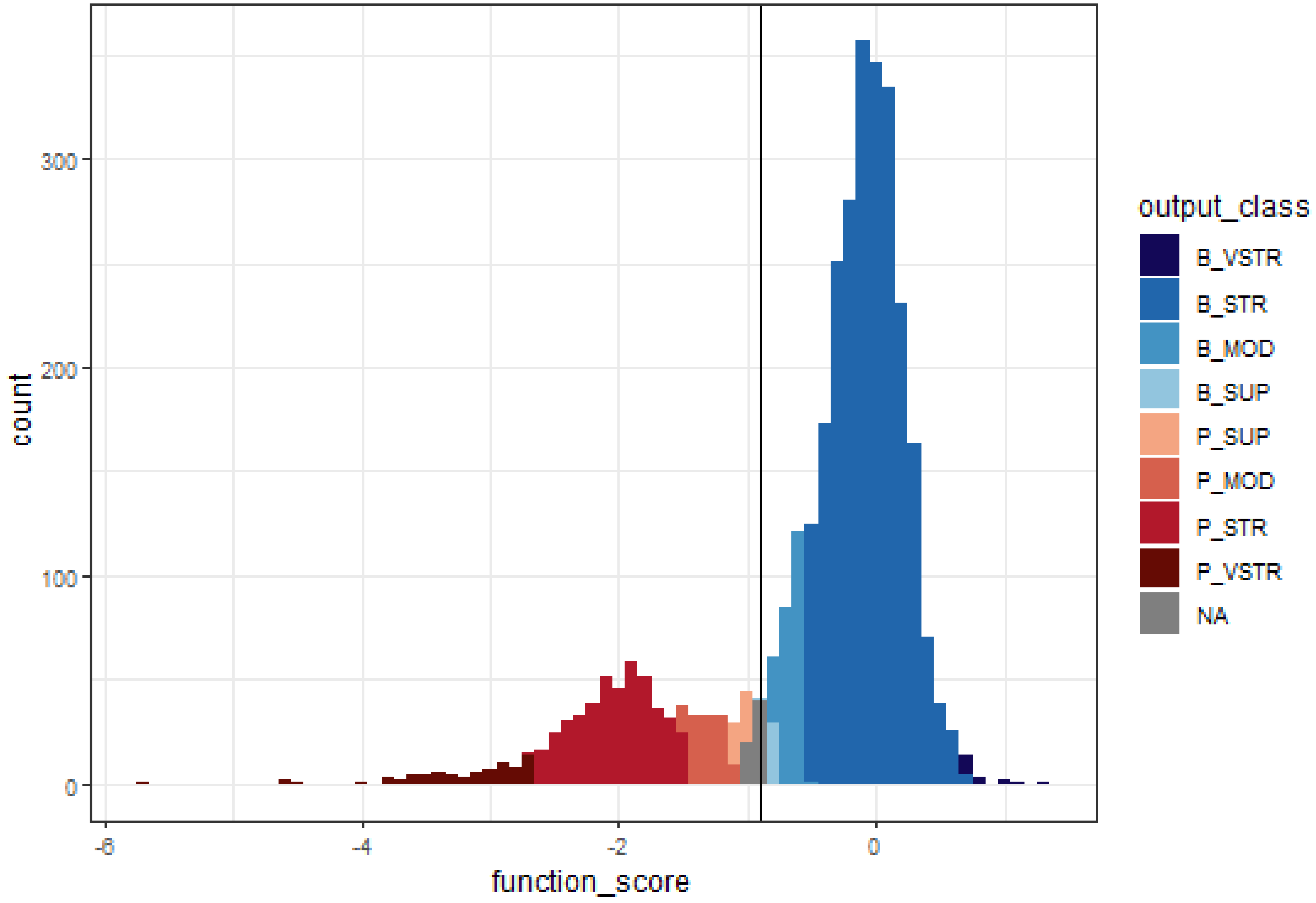


Example

2018 ClinVar dataset (all stars), Findlay et al 2018 BRCA1 dataset

Evidence Strength	Likelihood Odds (OddsPath)	Evidence points ($\log_{2.08}$[OddsPath])
P: Very Strong	350.41	+8
	168.44	+7
	80.98	+6
	38.93	+5
P: Strong	18.71	+4
	9.00	+3
P: Moderate	4.31	+2
P: Supporting	2.08	+1
B: Supporting	0.48	-1
B: Moderate	0.22	-2
B: Strong	0.05	-4
B: Very Strong	0.00	-8

Test data histogram and class



Objective 5: Learn how to **combine data from multiple assays**

Combining MAVEs for clinical classification

Charlie Rowlands

November 2025

Combination of MAVEs

BRCA1 c.5096G>C p.(Arg1699Pro)

Fernandes et al., 2019	
Functional class (fClass)	2
Functional class (fClass) category	Likely not pathogenic

Findlay et al., 2018	
Function score	-1.5485225389
Function probability	0.9997796775
Functional classification	LOF
Mean RNA score	-0.0154924365

Combining MAVEs with concordant results

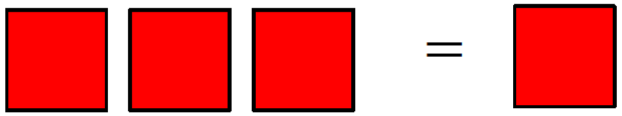
Option 1: Cap

Combined assay strength is same for both genes

Combined evidence strength is equal to the individual assay with the highest evidence strength

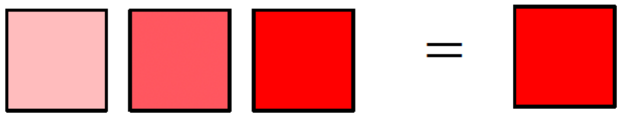
e.g. for Gene A:

MAVE 1	MAVE 2	MAVE 3
Strong	Strong	Strong



e.g. for Gene B:

MAVE 1	MAVE 2	MAVE 3
Supporting	Moderate	Strong



Alvaro Monteiro, ENIGMA meeting, 2024

Combining MAVEs with concordant results

Option 2: Stack

Combined evidence strength is calculated by summing the strengths of individual assays

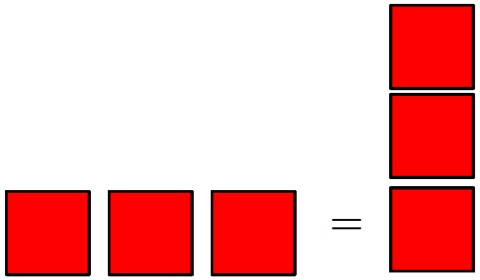
e.g. for Gene A:

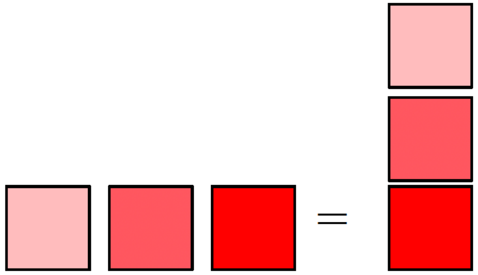
MAVE 1	MAVE 2	MAVE 3
Strong	Strong	Strong

e.g. for Gene B:

MAVE 1	MAVE 2	MAVE 3
Supporting	Moderate	Strong

Combined assay strength is greater for Gene A





Alvaro Monteiro, ENIGMA meeting, 2024

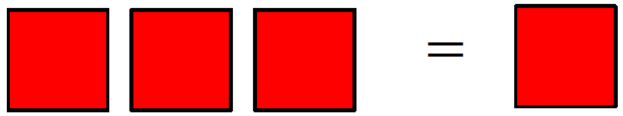
Combining MAVEs with discordant results

Option 1: Ignore

Variants with discordant MAVE scores remain as VUS

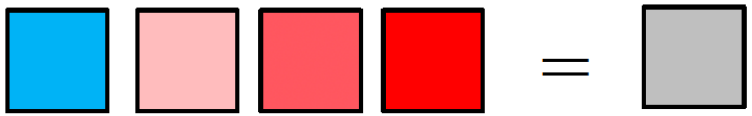
e.g. for Gene A:

MAVE 1	MAVE 2	MAVE 3
Strong	Strong	Strong



e.g. for Gene B:

MAVE 1	MAVE 2	MAVE 3	MAVE 4
Strong (benign)	Supporting (Path)	Moderate (Path)	Strong (Path)

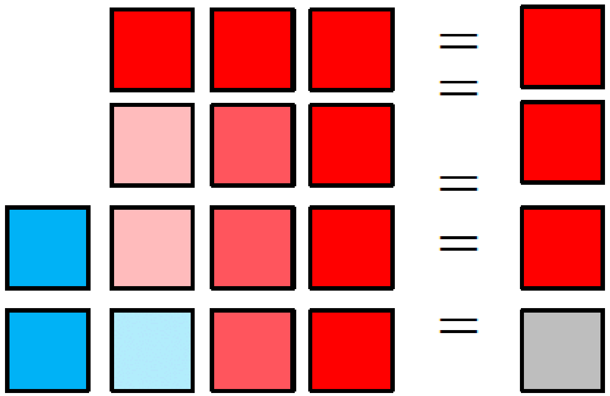


Alvaro Monteiro, ENIGMA meeting, 2024

Combining MAVEs with discordant results

Option 2: Combine

Discordant assay results for a variant are combined using the (3:1) preponderance of evidence rule



OR

Add up points corresponding to individual evidence strengths (as for “stacking” of concordant results)

Alvaro Monteiro, ENIGMA meeting, 2024

ENIGMA approach to *BRCA1/2* MAVE combination

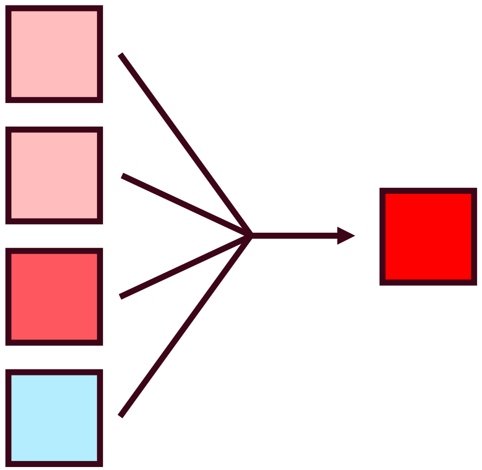
See **Specifications Figure 1C** for simplified flowchart/s to advise application of codes for functional data, in content of variant type and location within a (potentially) clinically important functional domain. Do not apply when conflicting results are present from well-established assays with sufficient controls, which cannot be explained by experimental design.

Using machine learning to combine MAVE scores

Option 3: Harmonise

Design computational classifier to take in individual scores and produce a single combined score

Evaluate performance of this new classifier in the same way as an individual functional study



Naïve Bayes classifiers

Construct Gaussian (normal) likelihood curves that best fit the pathogenic and benign score distributions across MAVEs of interest

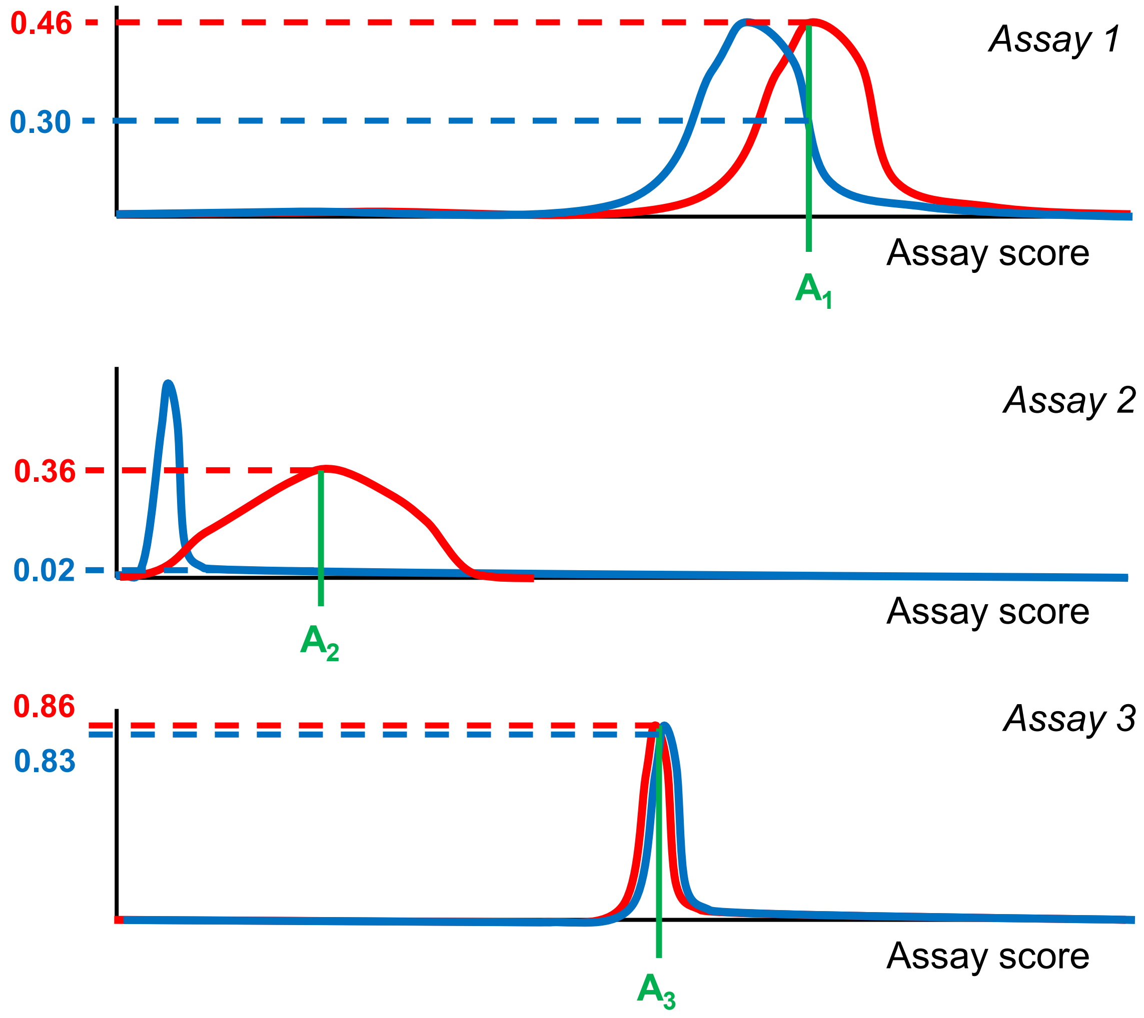
For a given variant, combine likelihood of pathogenicity and benignity across three assays and compare.

e.g. for **variant A**

$$P(A = Path|A_1, A_2, A_3) = 0.46 \cdot 0.36 \cdot 0.86 = 0.14$$

$$P(A = Ben|A_1, A_2, A_3) = 0.30 \cdot 0.02 \cdot 0.83 = 5.0 \times 10^{-3}$$

Likelihood that A is pathogenic is approximately 28.6 times greater than likelihood it is benign



Fayer et al., 2021. Combining MAVEs to resolve variants in *BRCA1*, *TP53* and *PTEN*

BRCA1

Findlay et al., 2018

Cell survival

TP53 (x4)

Giacomelli et al., 2019

- *Loss-of-function (etoposide selection)*
- *Loss-of-function (nutlin-3 selection)*
- *Dominant negative (nutlin-3 selection)*

Boettcher et al., 2019

Dominant negative (nutlin-3 selection)

Combine via Naïve Bayes classifier

PTEN (x2)

Matreyek et al., 2018

Protein abundance

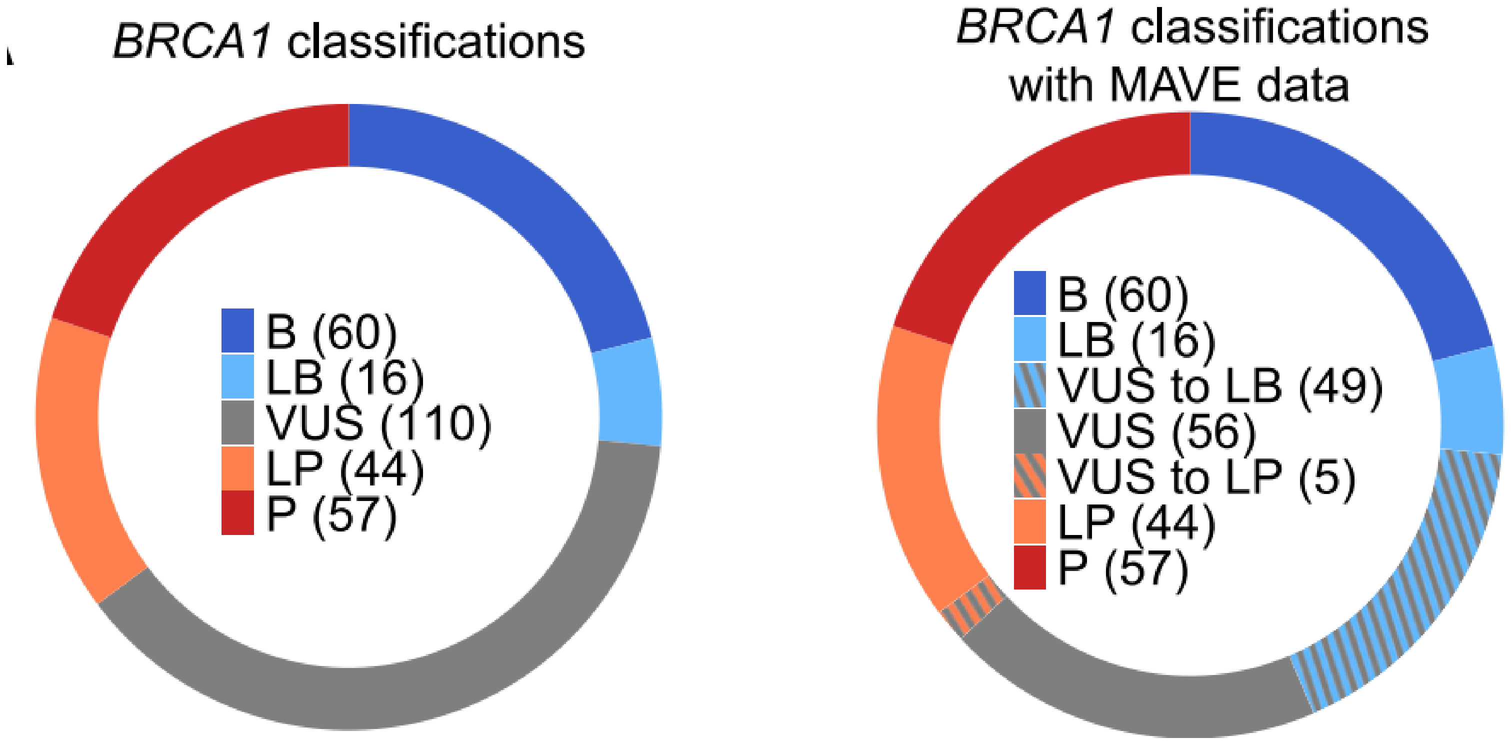
Mighell et al., 2018

Phosphatase activity

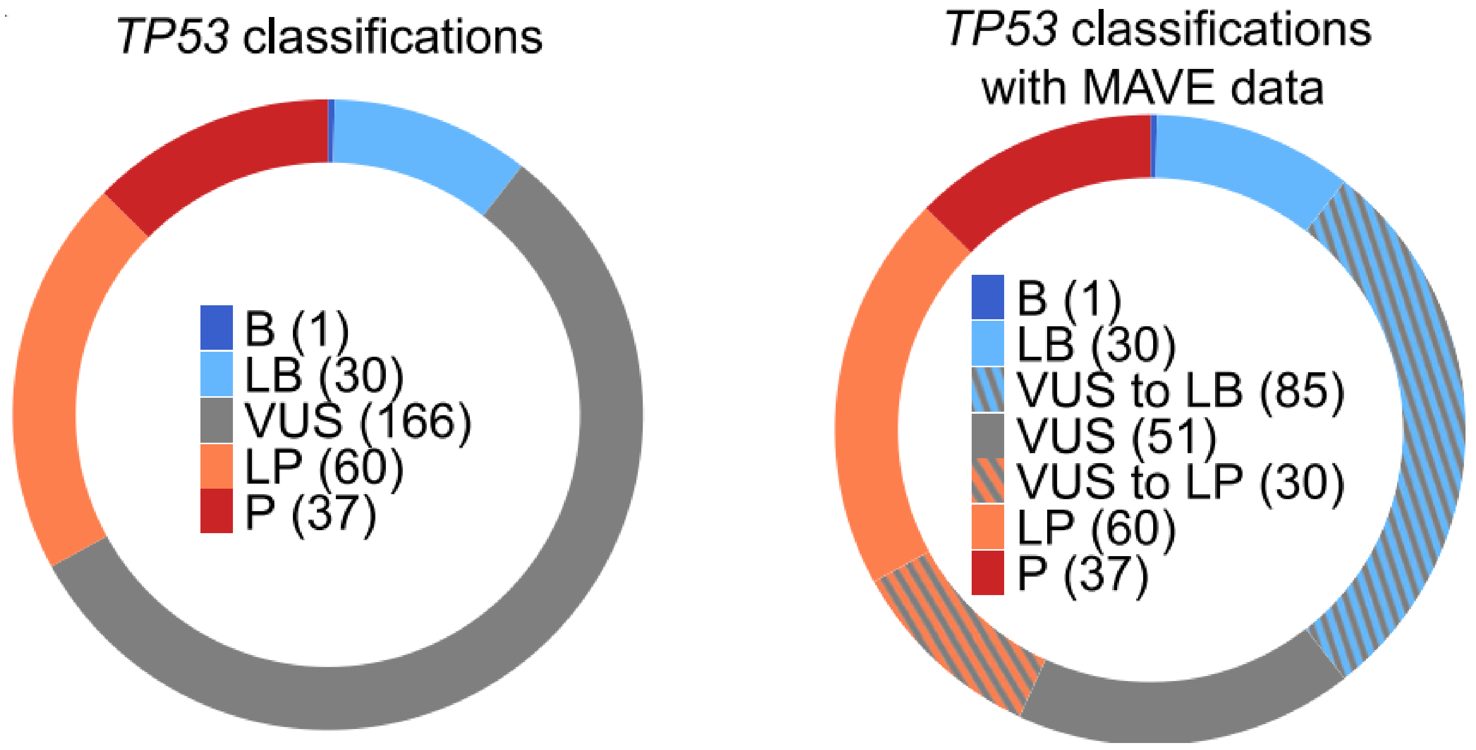
Too few benign missense variants to train classifier

→ use VCEP (expert group) guidelines on functional evidence integration

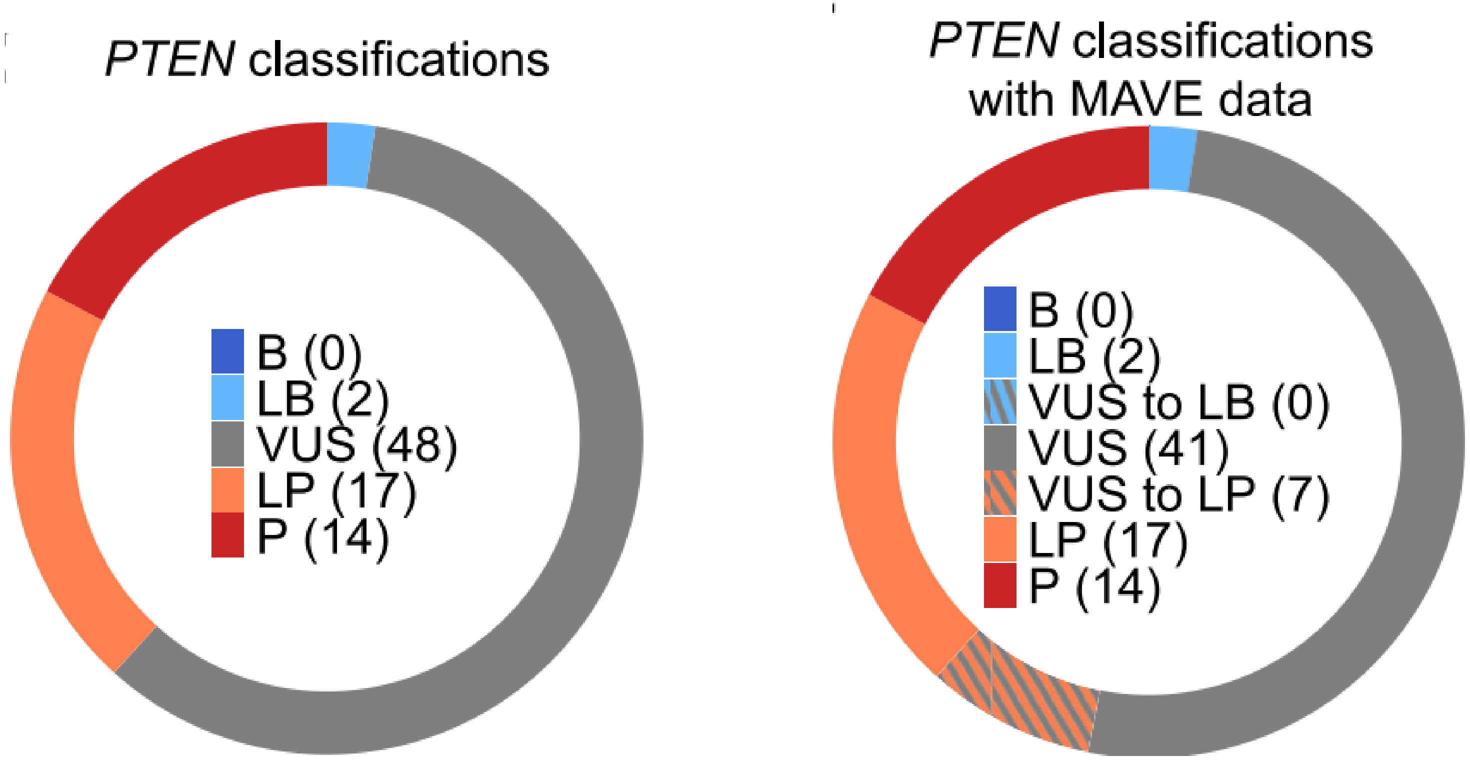
Fayer et al., 2021. Combining MAVEs to resolve variants in *BRCA1*, *TP53* and *PTEN*



Fayer et al., 2021. Combining MAVEs to resolve variants in *BRCA1*, *TP53* and *PTEN*

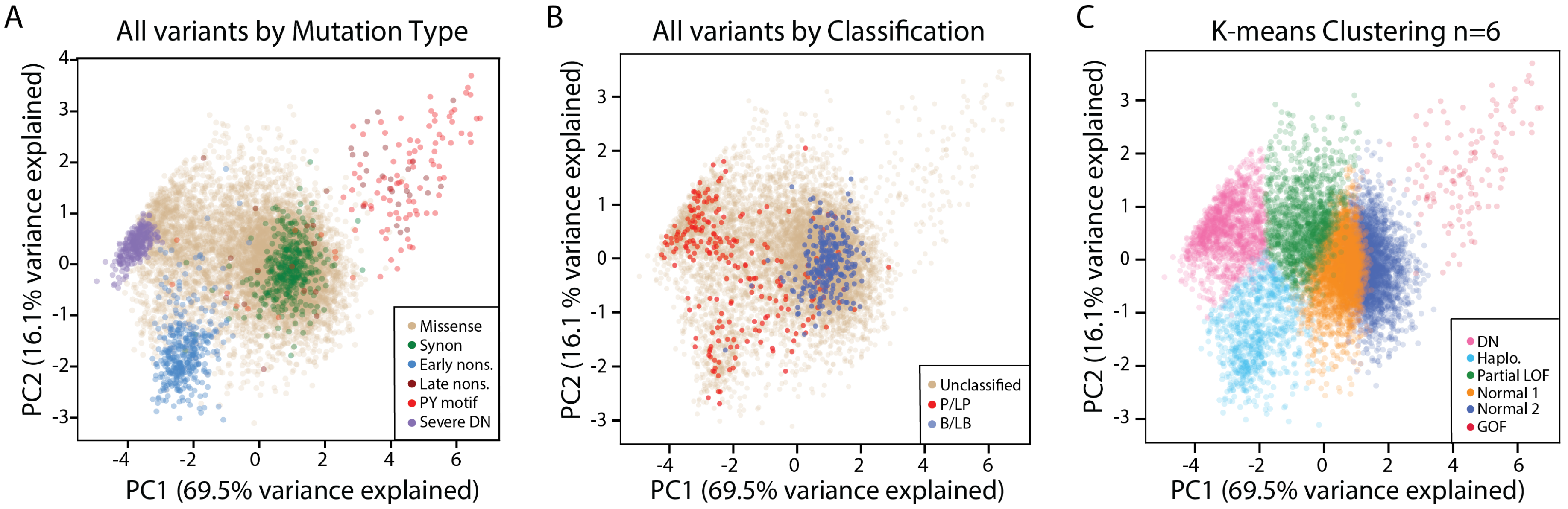


Fayer et al., 2021. Combining MAVEs to resolve variants in *BRCA1*, *TP53* and *PTEN*



Combining four MAVEs in *KCNQ1* via PCA and k-means clustering

- Variant-only abundance
- Variant-only function
- Heterozygous abundance
- Heterozygous function



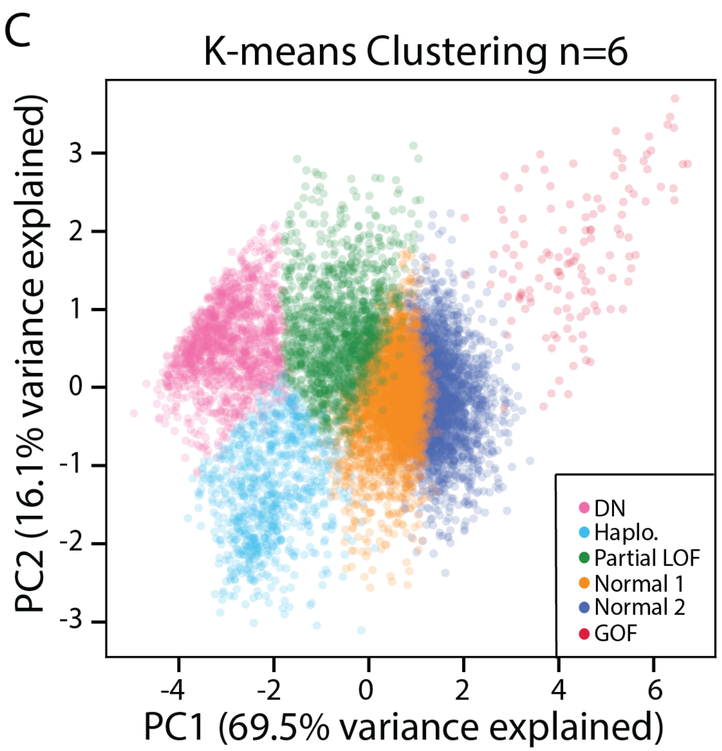
PCA distinguishes different variant mechanisms

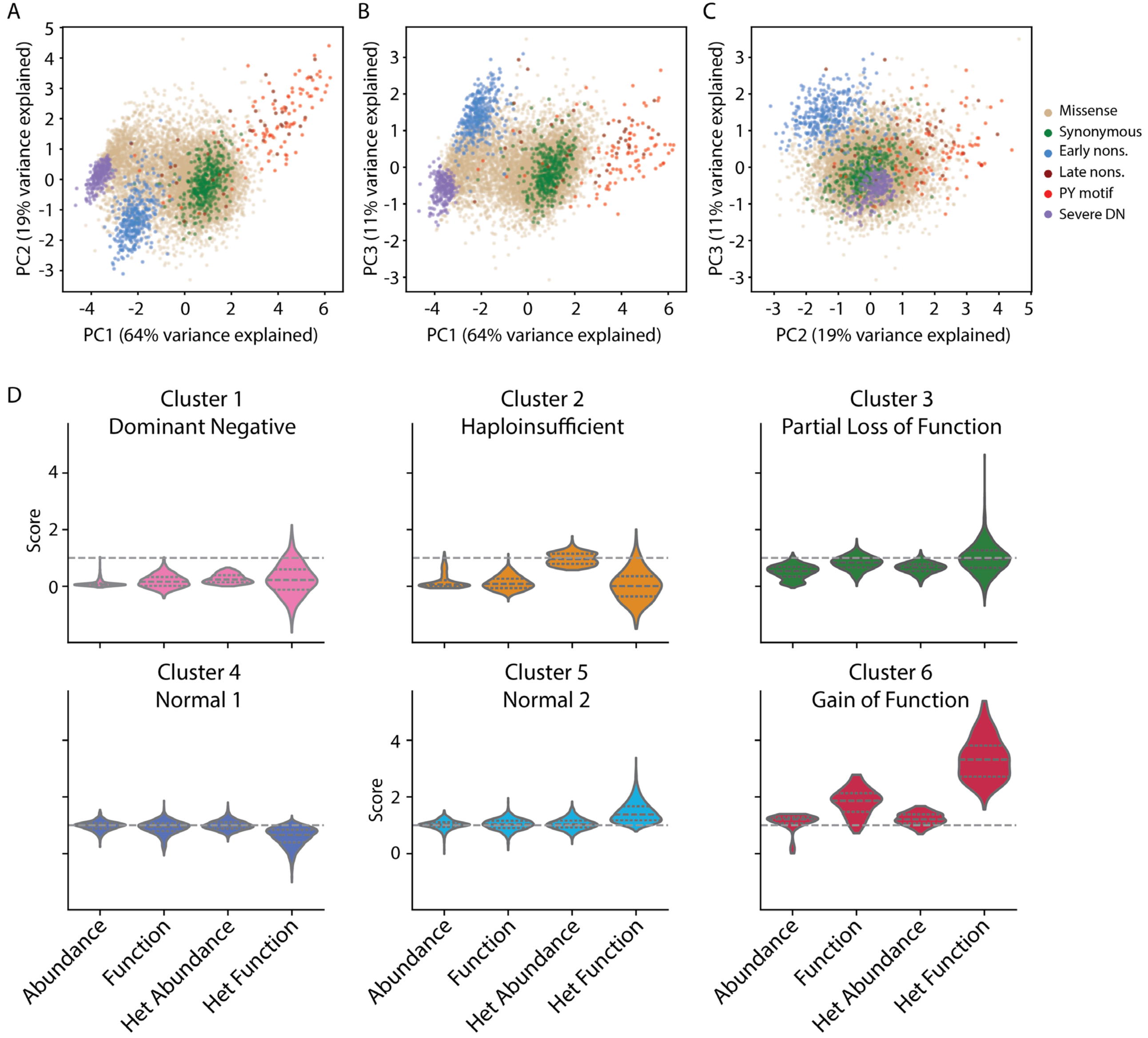
Many P/LP variants cluster with dominant negative variants

K-means clustering (n=6) best mapped to sensible groups of variants →clusters named based on manual inspection (next slide)

Andrew Glazer, unpublished

Combining four MAVEs in *KCNQ1* via PCA and k-means clustering





Andrew Glazer, unpublished

Comparing different approaches to MAVE combination

Combining five *TP53* MAVEs using three approaches

Questions?

Please contact wellcomeconnectingscience.org
for more information.